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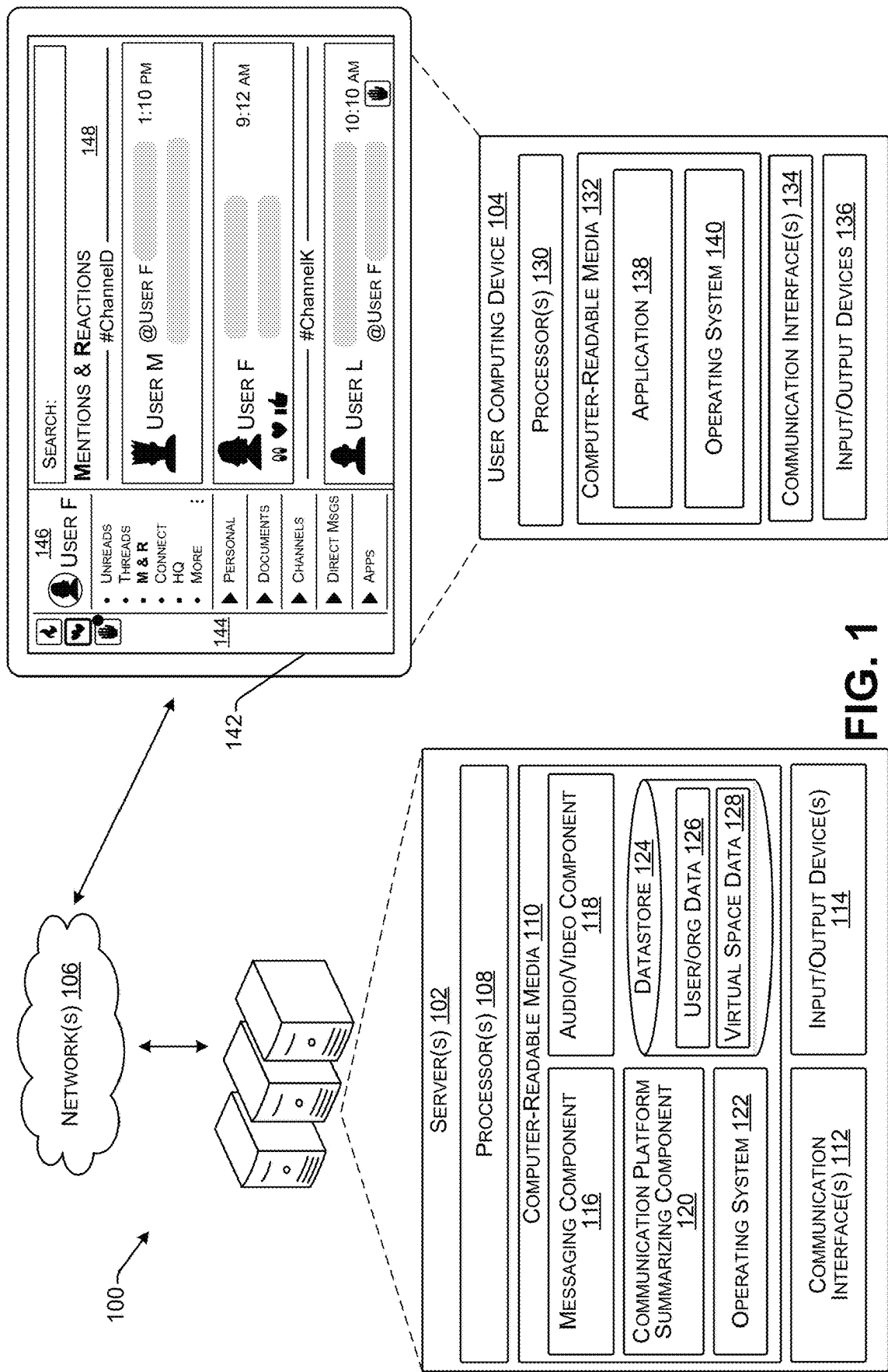
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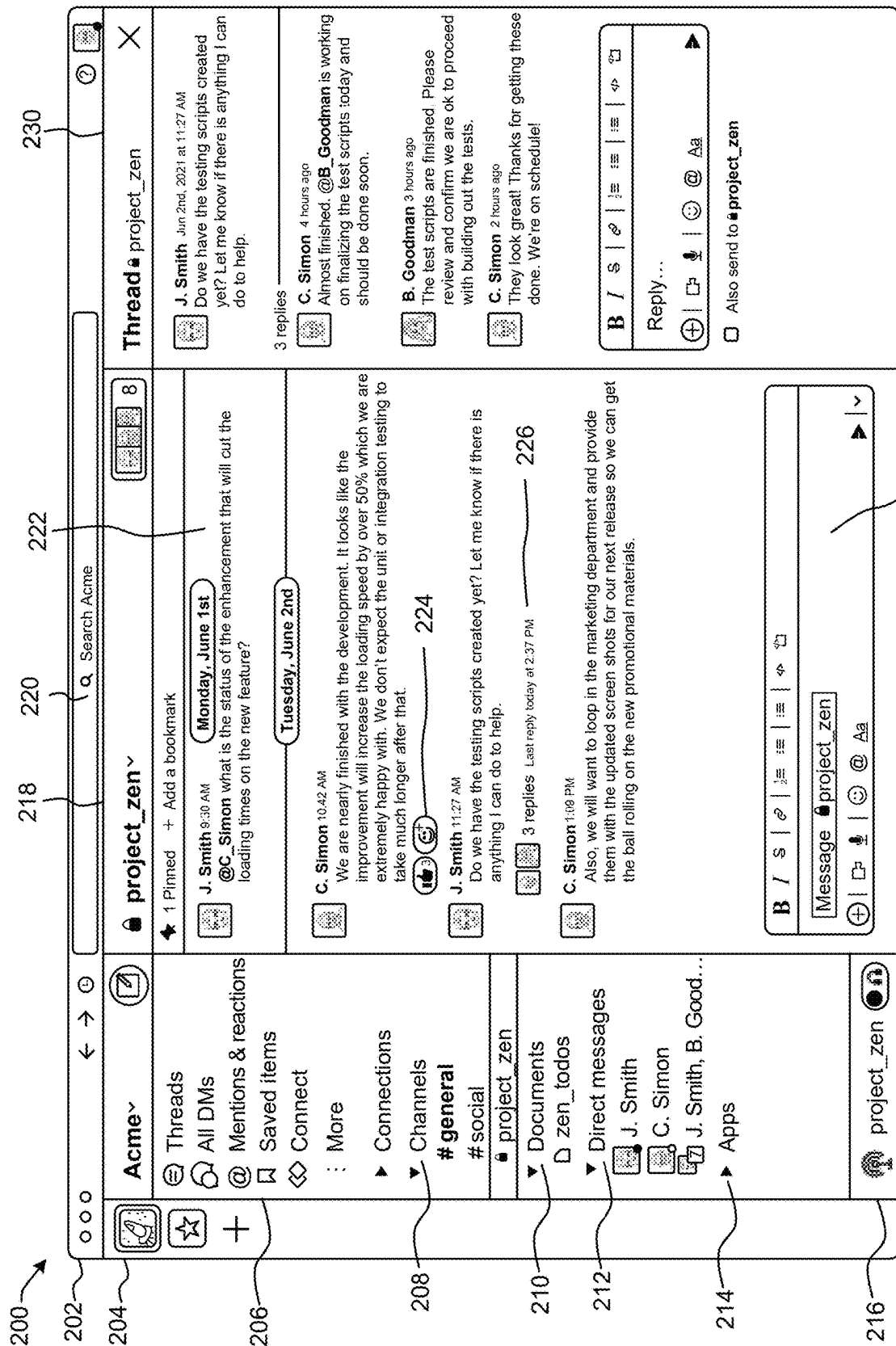
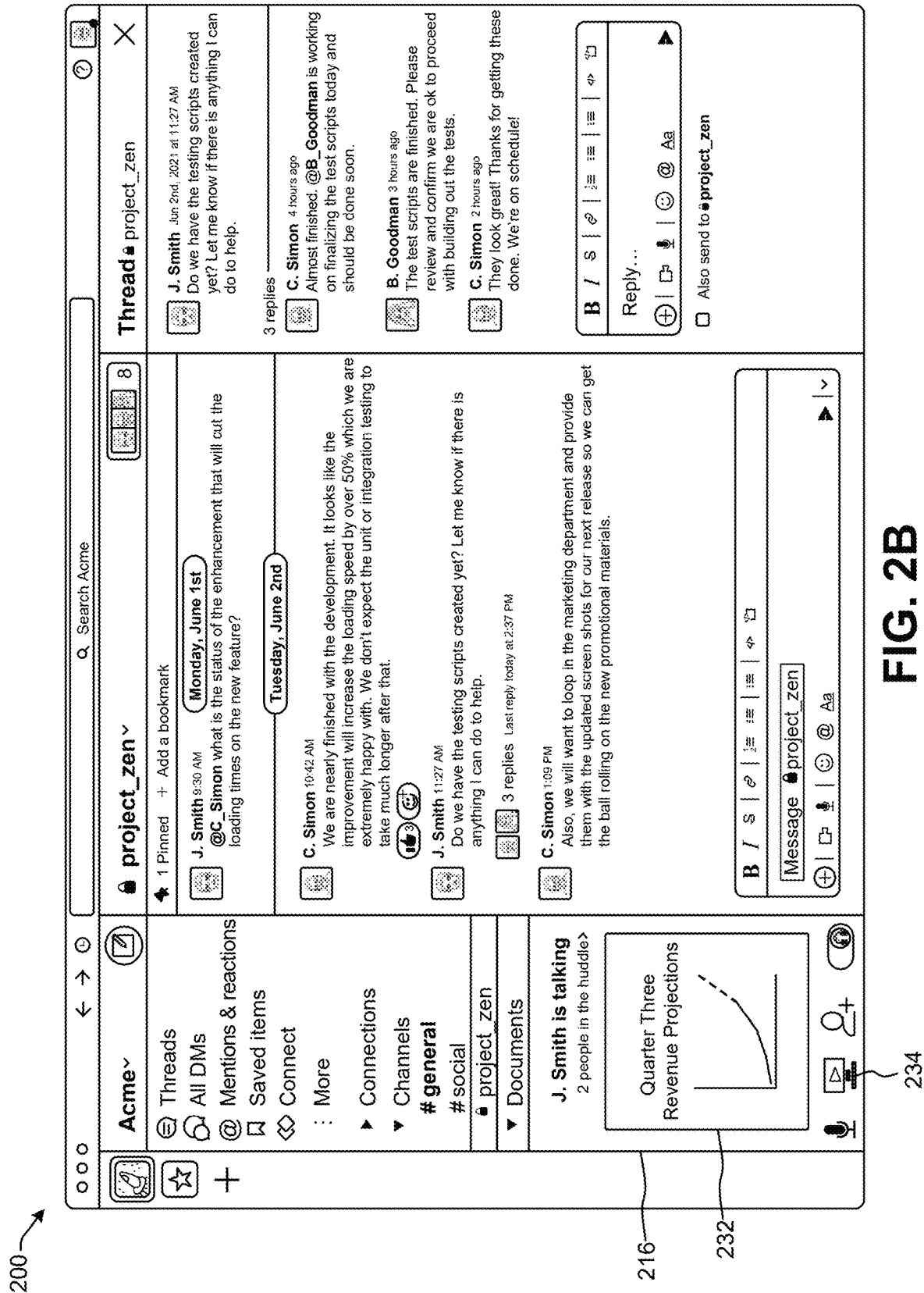
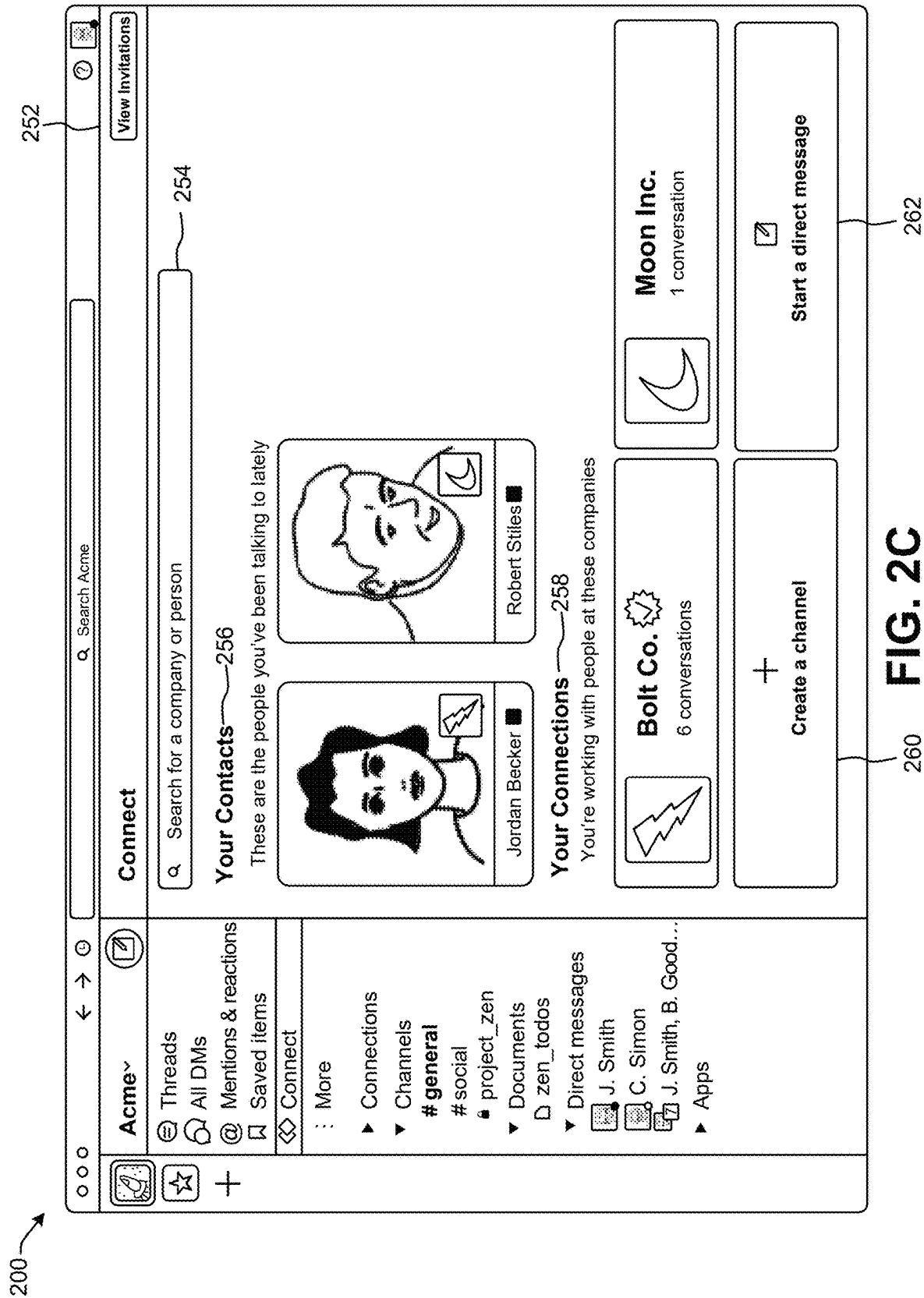


FIG. 2A





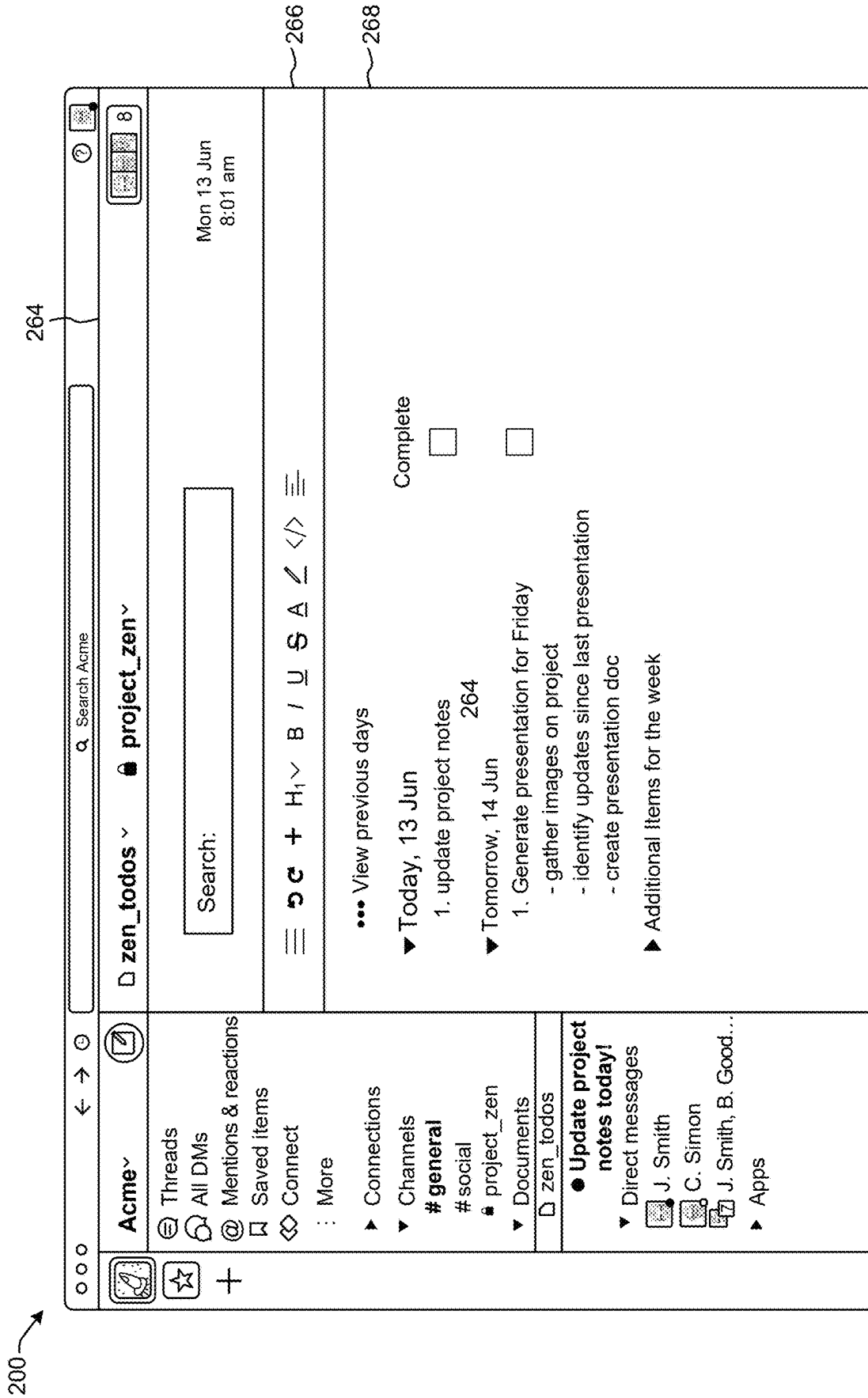


FIG. 2D

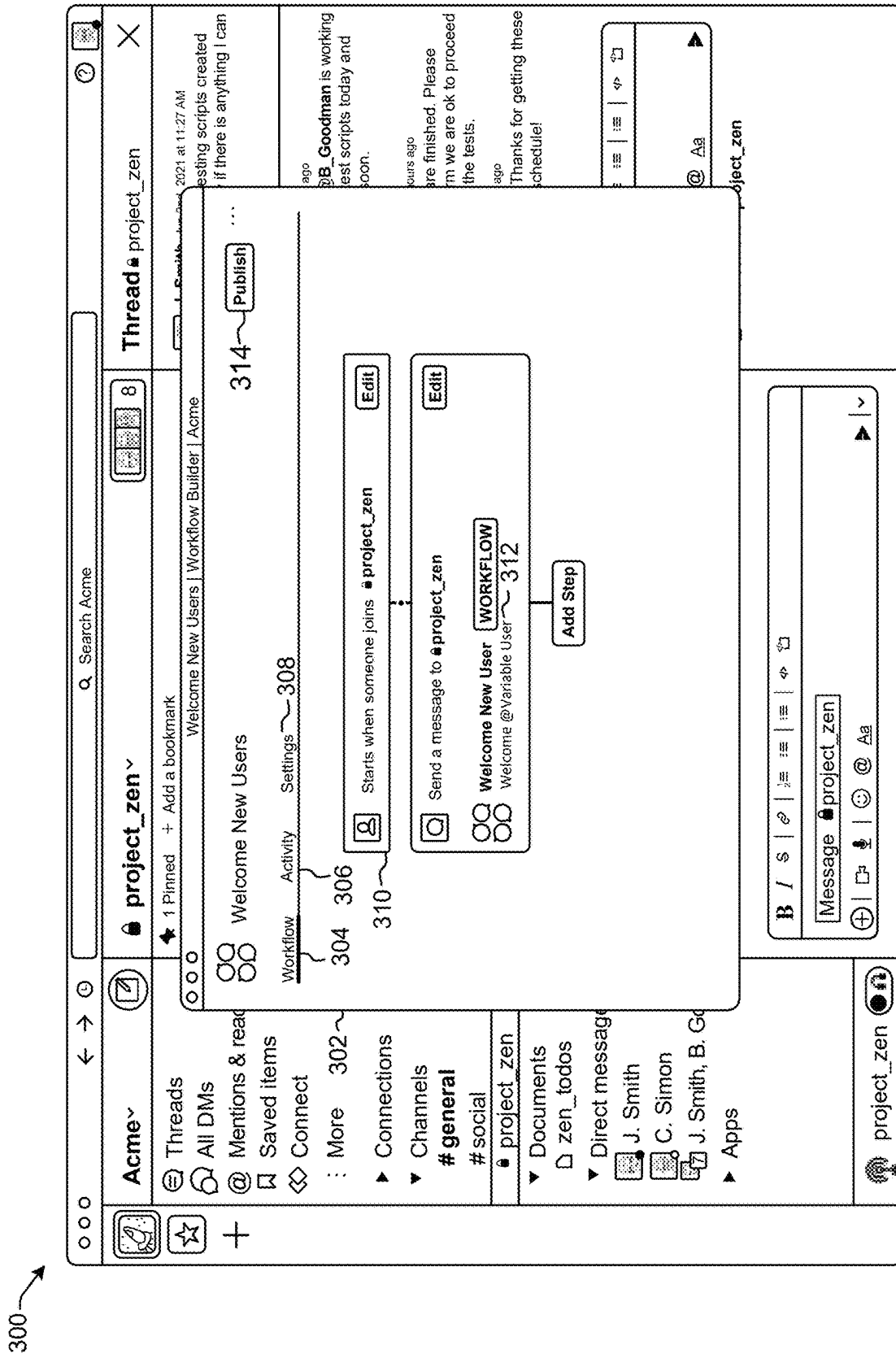


FIG. 3A

316

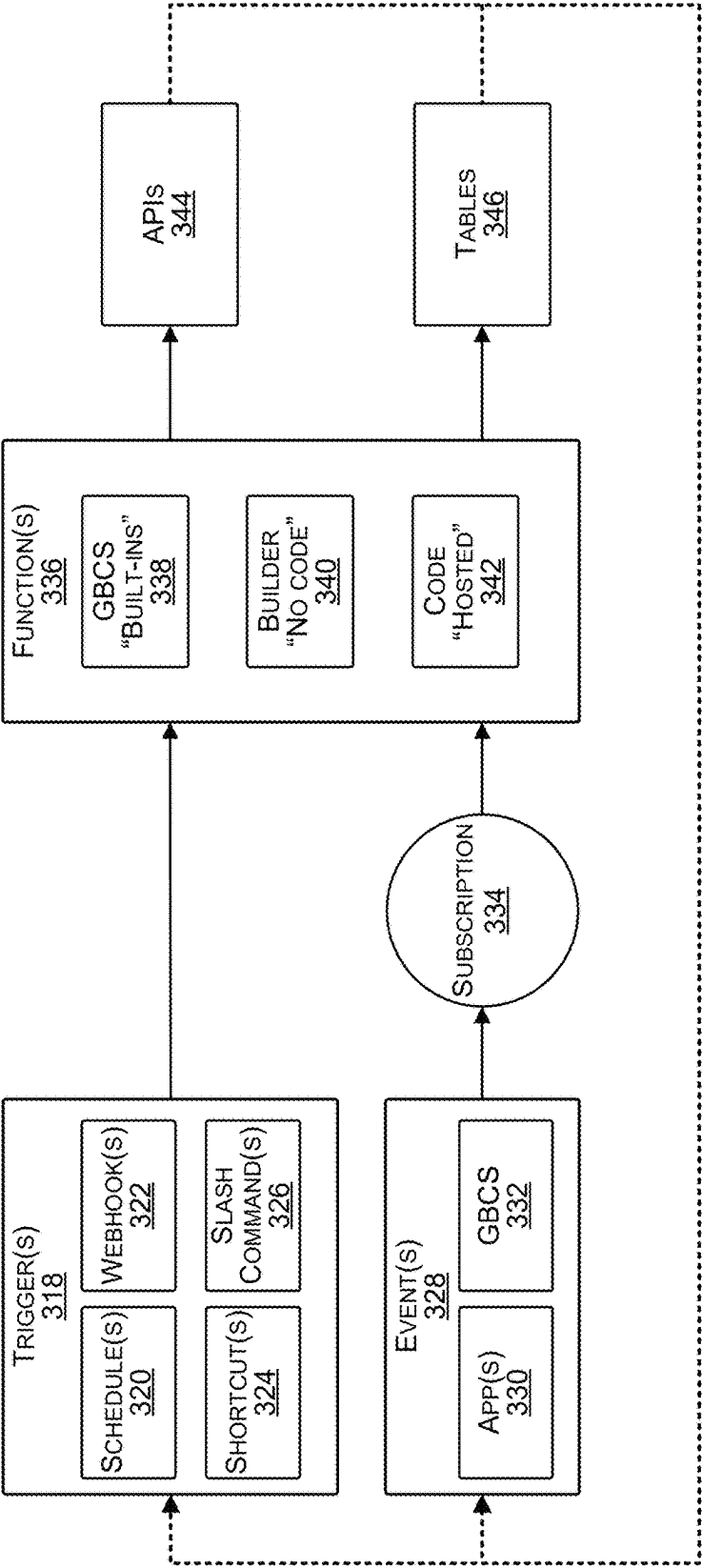


FIG. 3B

400 →

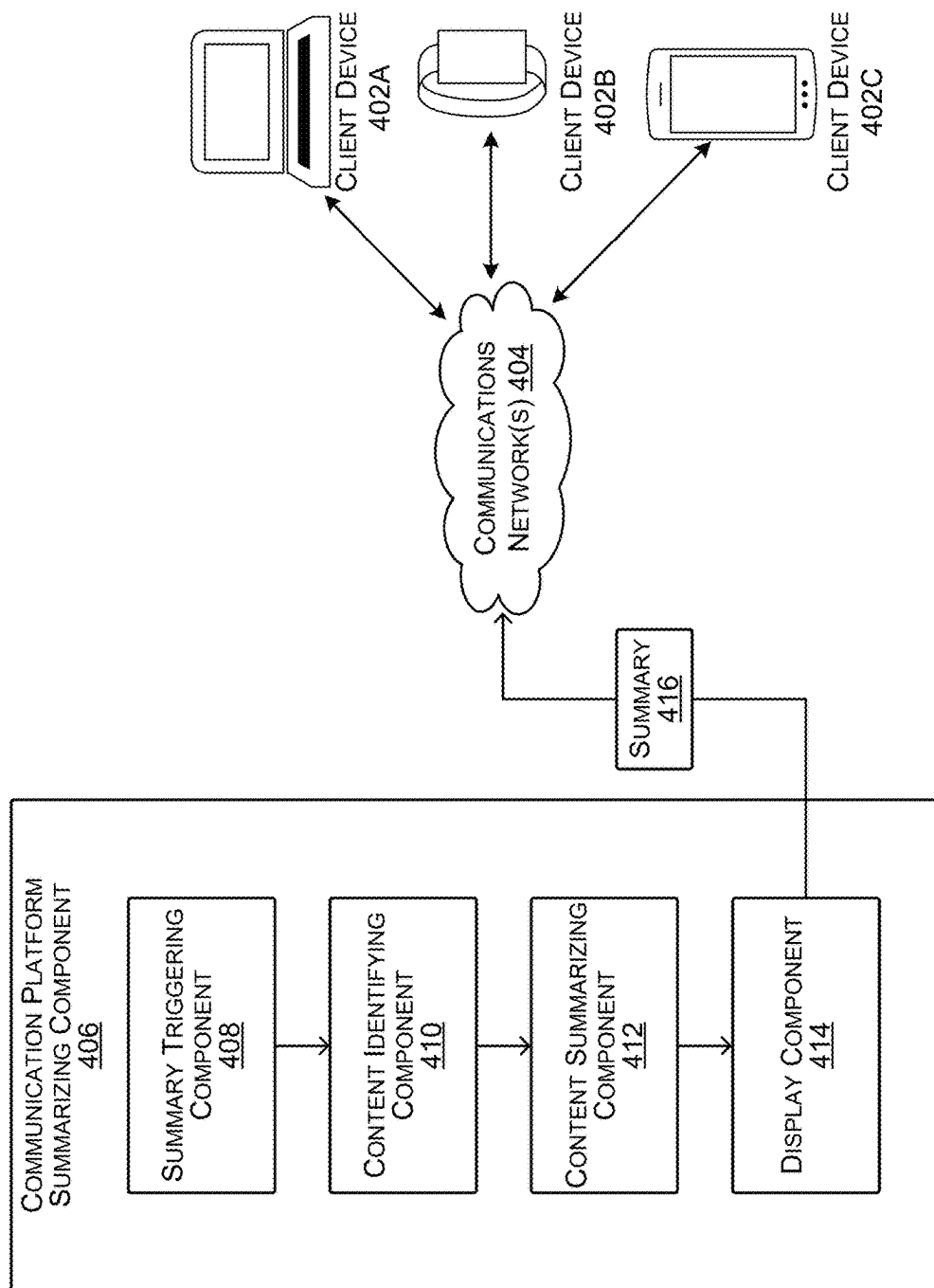
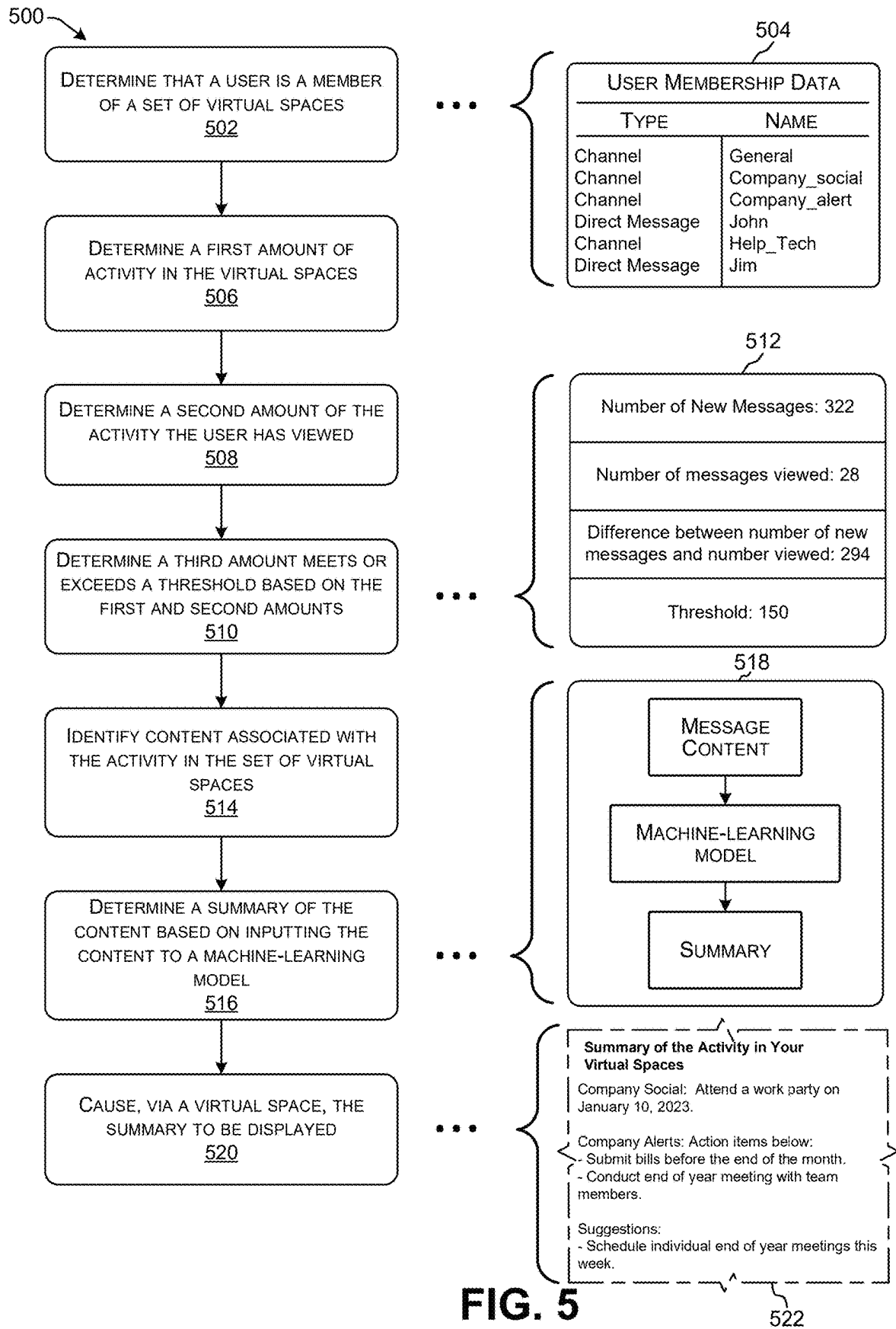


FIG. 4



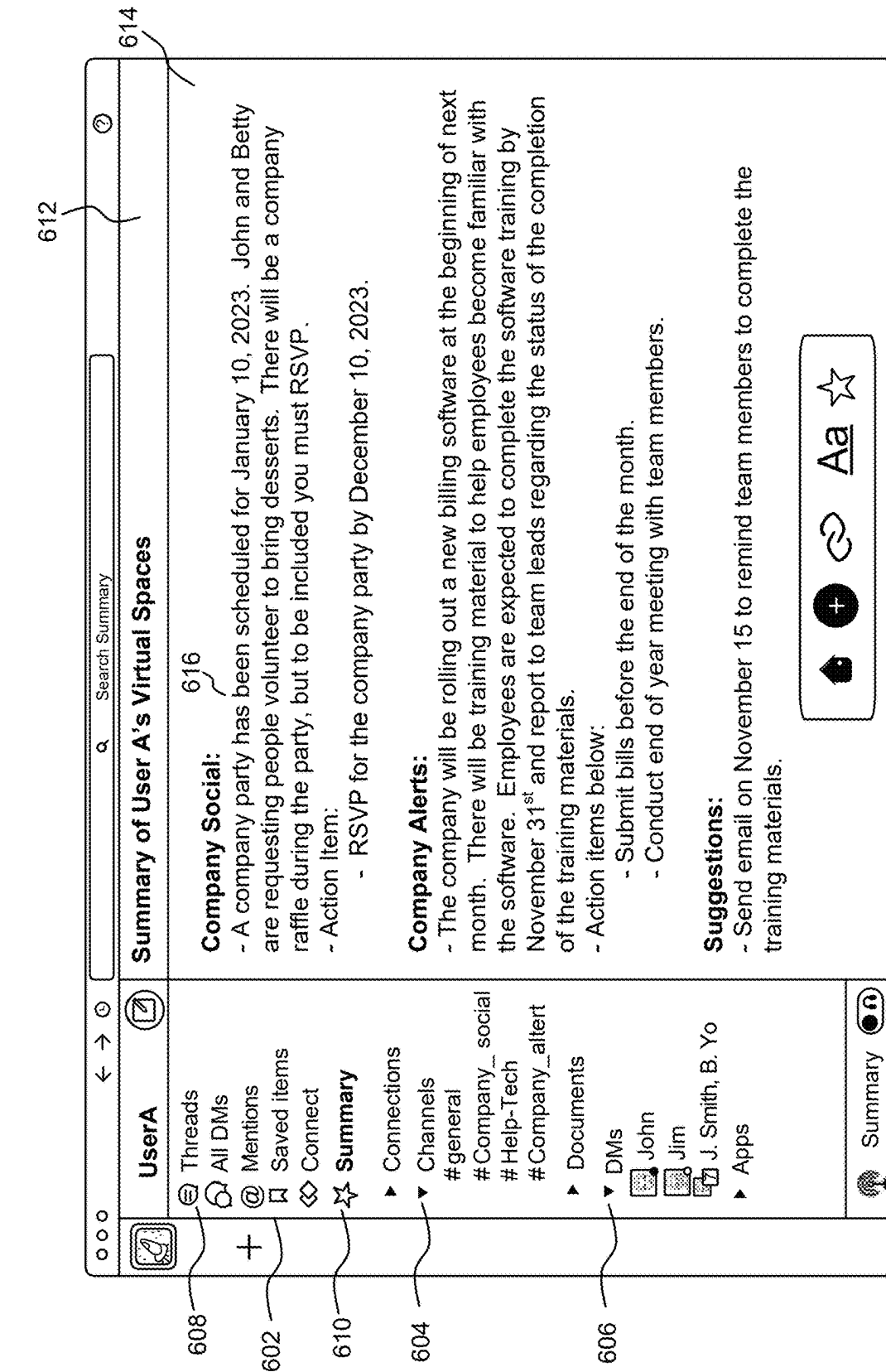


FIG. 6

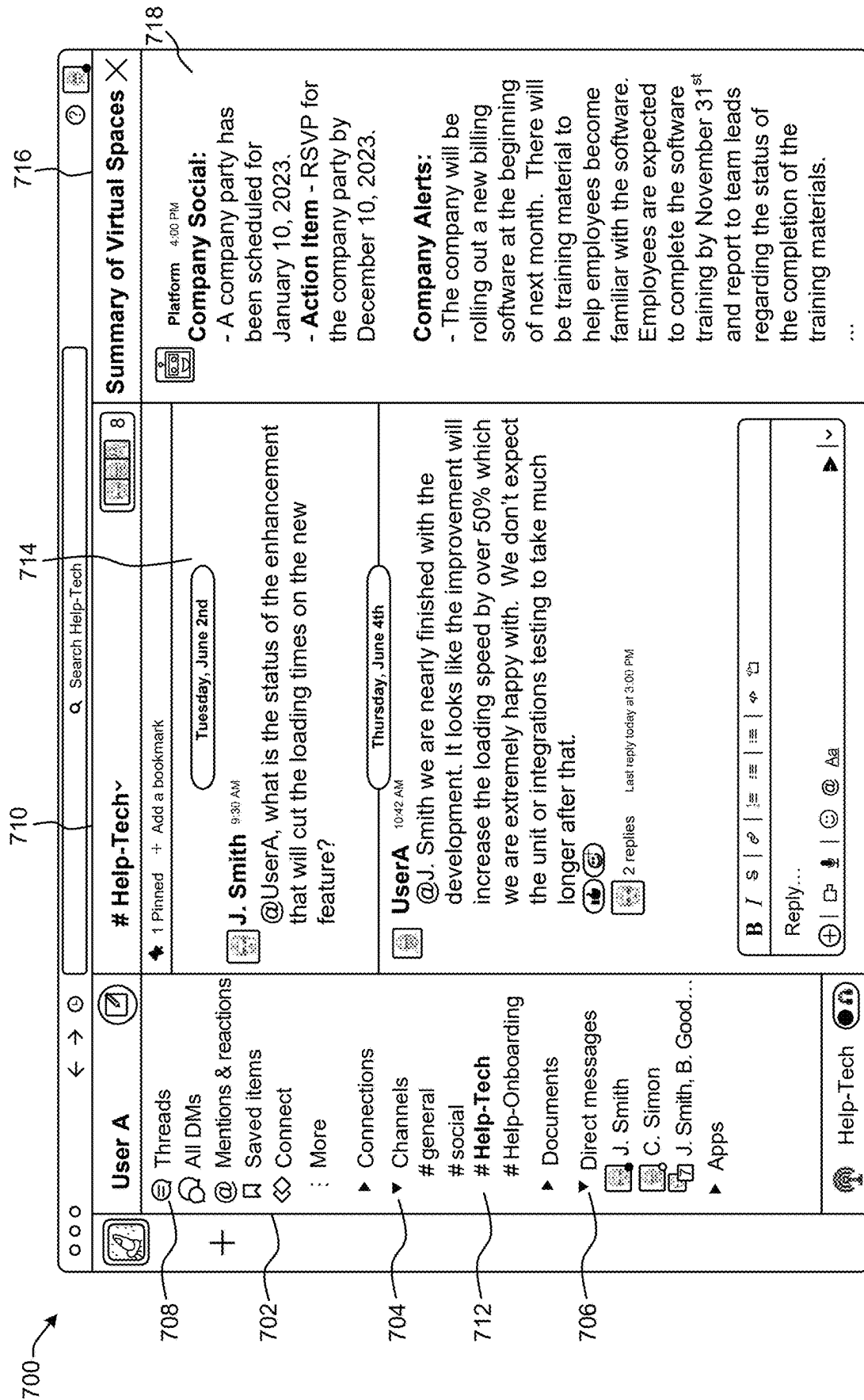


FIG. 7

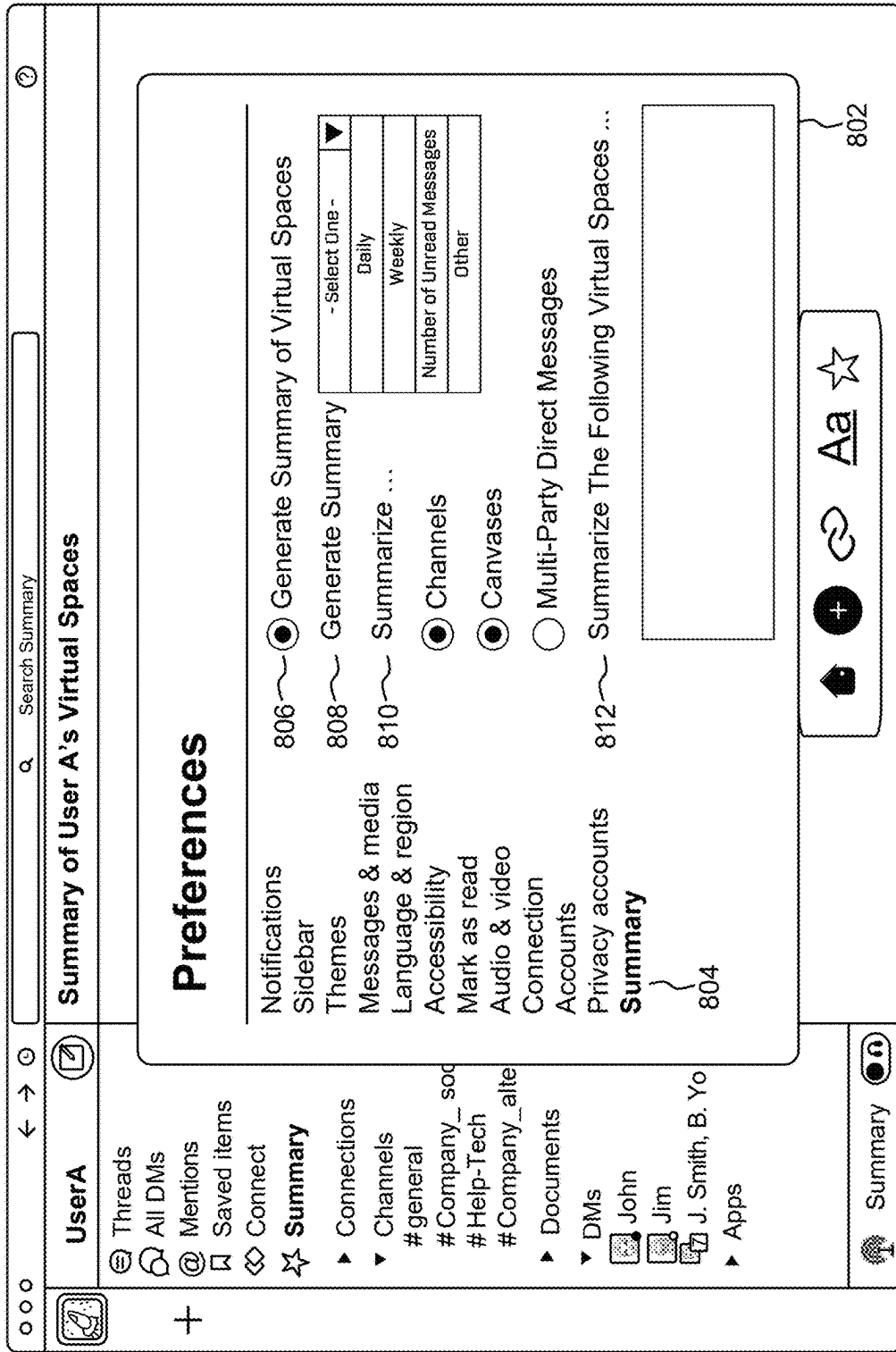


FIG. 8

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DISPLAYING A COMMUNICATION PLATFORM SUMMARY

TECHNICAL FIELD

Communication platforms are becoming increasingly more popular for organizations to facilitate work related communications. Users of such communication platforms can communicate with other users via channels, direct messages, an/or other virtual spaces. In some examples, a user may be a member or otherwise associated with multiple virtual spaces. Such virtual spaces may have various different purposes, such as serving as a company-wide general virtual space, a company social virtual space, a help desk/tech virtual space, and/or many other types of virtual spaces. In some examples, such virtual spaces may include large quantities of members who may contribute to or otherwise post to the virtual spaces. In some examples, in order to maintain an up-to-date understanding of the updates and/or discussions in the virtual spaces, the user may be required to access and/or view each individual virtual space. In some examples, accessing and/or viewing such a large volume of virtual spaces and/or content in this manner may be inefficient and time consuming to users. This may be especially true in the case where there is a voluminous number of posts and/or questions in virtual space, resulting in users spending excessive amounts of time attempting to identify the valuable contents within each virtual space.

BRIEF DESCRIPTION OF THE DRAWINGS

The detailed description is described with reference to the accompanying figures. In the figures, the left-most digit of a reference number identifies the figure in which the reference number first appears. The use of the same reference numbers in different figures indicates similar or identical components or features. The figures are not drawn to scale.

FIG. 1 illustrates an example system for performing techniques described herein.

FIG. 2A illustrates a user interface for a group-based communication system for certain examples.

FIG. 2B illustrates a user interface for multimedia collaboration sessions within the group-based communication system for certain examples.

FIG. 2C illustrates a user interface for inter-organization collaboration within the group-based communication system for certain examples.

FIG. 2D illustrates a user interface for collaborative documents within the group-based communication system for certain examples.

FIG. 3A depicts a user interface for workflows within a group-based communication system.

FIG. 3B depicts a block diagram for carrying out certain examples, as discussed herein.

FIG. 4 depicts an example block diagram illustrating the interactions of components of a communication platform summarizing component configured to generate and/or display summarized virtual space data.

FIG. 5 is a pictorial flow diagram illustrating an example process for determining whether to generate a summary for a user, identifying content associated with a set of virtual spaces, generating a summary of the content, and displaying the summary via a user interface of the user.

FIG. 6 illustrates an example user interface associated with a communication platform, as described herein, for displaying a summary of recently posted content to a summary virtual space.

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FIG. 7 illustrates an example user interface associated with a communication platform, as described herein, for displaying a summary of recently posted data to a side pane.

FIG. 8 illustrates an example user interface associated with a communication platform, as described herein, for allowing a user to determine preferences about when and/or how the communication platform generates a summary of the content associated with a set of virtual spaces.

DETAILED DESCRIPTION

As discussed above, conventional techniques for a user to catch up on recently posted content across a set of virtual spaces may be suboptimal and can result in the user spending excessive amounts of time manually attempting to identify the valuable contents in each virtual space.

Techniques for generating and/or displaying a summary relating to content recently posted in multiple virtual spaces are discussed herein. As described herein, upon identifying that a user is a member of a set of virtual spaces, machine-learning models may be used to generate a summary of content recently posted across the set virtual spaces. In some examples, a communication platform may determine whether to generate and/or display a summary of the recently posted content (e.g., messages, content items, updates, files, etc.) in a set of virtual spaces. For instance, the communication platform may generate a summary based on analyzing user activity data, message data, and/or virtual space data corresponding to a user. That is, the communication platform may determine or otherwise identify a set of virtual spaces that the user is a member of. Spanning a pre-determined period of time, the communication platform can determine a first number of content items (e.g., messages, updates, etc.) that have been posted to the set of virtual spaces. The communication platform may also determine, over the same period of time, a second number indicating the number of the content items the user has viewed and/or accessed. Based on the first and second numbers, the communication platform may determine a third number that indicates how many of the content items the user has yet to view. Based on the third number meeting or exceeding a threshold value, the communication platform may determine that it should generate a summary for the user. In such instances, the communication platform may generate a summary of the content that has been posted to the set of virtual spaces and cause the summary to be displayed via a user interface of the user. As discussed throughout the disclosure, the techniques may improve user experience by enabling users to receive relevant information regarding the recently posted content without the user being required to access each individual virtual space to get “caught up” on the decisions and/or discussions being had in the virtual space. Such techniques may reduce processor cycles that may be performed to facilitate the user navigating to each individual virtual space.

When helping a user get “caught up” in a virtual space, conventional techniques may be ineffective and/or disrupt the user experience. In some examples, a user may be a member of a large number of virtual spaces (e.g., channel(s), board(s), direct messaging instance(s), multiparty direct messaging instance(s), etc.). Further, such virtual spaces may include large quantities (e.g., 10, 100, 1,000, 10,000, etc.) of other users that are also members of the virtual spaces. Such users may be able to contribute to the virtual spaces by posting and/or reacting to messages. Over time, some or all of the virtual spaces may receive many new posts and/or reactions to such posts. However, in some examples,

a user may be out of office (e.g., vacation, sick, personal matter, etc.) and/or busy working on a different project. In such instances, upon returning to office and/or completing the project, a user may identify or otherwise notice the large quantities of recently posted content across the user's various virtual spaces that the user has yet to view. If the user wants to get "caught up" on the virtual spaces, conventional systems may cause the user to have to access and/or view each virtual space. However, accessing and/or reading the recently posted messages from each virtual space may be time consuming and difficult, and may require the consumption of a significant amount of computing resources. Consequently, fully manual reading of recently posted virtual space content can be time consuming, inefficient, and error-prone technique for getting "caught up" on the many virtual spaces that the user is a member of.

To address these and other technical problems and inefficiencies, the systems and/or techniques described herein may include a communication platform summarizing component (which also may be referred to as a "summarizing component" or a "virtual space summarizing component") configured to generate and/or display a summary of the recently posted content across multiple virtual spaces. The technical solutions discussed herein solve one or more technical problems associated with conventional techniques resulting in a suboptimal user experience.

Initially, the communication platform may determine whether to generate a summary for a user profile. A communication platform can be a group-based communication platform, a channel-based messaging platform and/or any other platform for facilitating communication between and among users. Users can use a variety of devices (or "user devices") to access the communication platform. Such devices may include any suitable type of computing device, e.g., portable, semi-portable, semi-stationary, or stationary. In some examples, the communication platform can determine whether to generate a summary based on a user request and/or the number of unread, recently posted messages across multiple virtual spaces meeting or exceeding a threshold value.

For instance, a user may request that the communication platform generate a summary of the content posted to some or all virtual spaces of which the user (or user profile) is a member. In some examples, the user may request the summary in a variety of ways, such as by selecting an object in the virtual space and/or by modifying user preferences corresponding to a user profile of the user. The communication platform may include a virtual space that includes a list of one or more user preferences controlling the ways in which the communication platform functions or otherwise operates. Some user preferences may include a notification preference, a sidebar preference, a theme preference, a message and media preference, a language preference, an account preference, a privacy accounts preference, a summary preference, and/or any other type of preference. In some instances, the user may navigate to the user preferences virtual space and select the "summary" preference settings. In such instances, the communication platform may display an overlay interface, popup box, and/or any other type of interface configured to enable the user to modify (e.g., customize) the summary preferences as applied to the user profile. The summary preference interface may include a plurality of input fields allowing the user to customize how and/or when the communication platform is to generate and display a summary. Examples of the input fields may include a condition that, upon the being satisfied, causes the communication platform to generate a summary (e.g., daily,

weekly, number of unread messages, etc.). Further other input fields may allow the user to specify which virtual spaces to summarize (e.g., channels, boards, canvases, etc.). As such, the user may modify the summary preference settings which may send a request to the communication platform to generate a summary according to the preferences listed therein.

Alternatively or additionally, the communication platform may determine whether to generate a summary based on the communication platform identifying a threshold number of messages that have yet to be viewed by a user across the user's multiple virtual spaces. That is, the communication platform may infer when a summary of the virtual spaces would be most helpful to the user. The communication platform may make such a determination based on virtual space and/or user activity data. For example, the communication platform may identify a set of virtual spaces with which the user profile is a member (e.g., based on membership data associated with the user profile). The communication platform may determine the set of virtual spaces based on analyzing the user profile virtual space membership data corresponding to the user profile.

Based on identifying the set of virtual spaces, the communication platform may determine, for a period of time, the number of posts (e.g., messages, content items, workflows, reactions, images, GIFs, updates, files, lists, documents, emojis, etc.) that have been displayed via the virtual spaces. That is, the communication platform may analyze some or all of the virtual spaces in the set and determine a number of posts that members of the virtual spaces (or other users having user profiles with permissions sufficient to post in such virtual spaces) have made in the virtual space.

In some examples, the communication platform may determine, over the same period of time, a number of posts (from the same set virtual spaces) that the user (or user profile) has yet to view. For instance, the communication platform may analyze user activity data of the user profile to determine the number of posts that have been displayed in the virtual spaces which the user has viewed. A user may view a post by accessing the virtual space and/or interacting with the post in some way. In some examples, the communication platform can determine the number of posts that the user has yet to view based on comparing (e.g., subtracting) the number of posts made to the virtual spaces with the number of such posts the user profile has viewed. In some examples, if the number of posts yet to be viewed meets or exceeds a threshold, the communication platform may determine that there is a sufficient amount of data (e.g., information) that the user has yet to consume and as such, the communication platform may determine that it may be helpful for the user to receive a summary of the content posted to the set of virtual spaces. As noted above, the summary may enable the user to get "caught up" with the recently posted content without having to access and/or view each individual virtual space.

Based on determining that the communication platform may generate a summary for the user, the communication platform may identify content to summarize. Such content may be associated with the set of virtual spaces and/or posts contained therein. That is, for the communication platform to summarize the content recently posted in the set of virtual spaces, the communication platform may identify data corresponding to the set of virtual spaces to summarize. In some examples, the communication platform may prioritize the content to summarize, such as by prioritizing the virtual spaces and/or the individual messages.

That is, the communication platform may prioritize or otherwise rank which messages are of the highest importance to summarize. Specifically, the communication platform may rank the messages based on which content is the most relevant to the user receiving the summary. In some examples, the communication platform may rank some or all of the messages across some or all of the virtual spaces of which the user is a member, or as according to the user request. The messages can be ranked based on a variety of factors, such as the content of the messages, a role of the user that posted the message (e.g., a message posted by an organization leader may be highly ranked, a message posted by a subordinate employee may be ranked lower than the message posted by the organization leader, etc.), a role of the user receiving the summary (e.g., certain messages can be relevant to the user based on their role), a relationship between the posting user and the summary receiving user (e.g., organization chart), interaction data between the posting user and the summary receiving user (e.g., a frequency of interactions, a recency of interactions, a purpose of the interaction, etc.), interaction data between the summary receiving user and the virtual space within which the messages are posted, and/or any other factor.

Additionally or alternatively, the communication platform can rank which virtual spaces are of the highest importance to summarize. Specifically, the communication platform may rank the virtual spaces based on which virtual spaces are the most relevant to the user receiving the summary. The virtual spaces may be ranked based on a variety of factors, such as the activity level of the user in the virtual space (e.g., a higher activity level can indicate that the virtual space is of higher importance to the user), a purpose of the virtual space to the user and/or the organization the user belongs to (e.g., company-wide channel may be of high importance to the user, a meme channel may be of lesser importance to the user, etc.), a number of unread posts in the virtual space, types of content items in the virtual space, and/or updates in the virtual space (e.g., a higher number of unread content may be of higher importance to summarize since it may take more time for the user to “catch up”), recency data indicating a period of time since the user joined the virtual space (e.g., a virtual space the user recently joined may be highly important to summarize since there may be large amounts of content to “catch up” on), and/or any other factor. In some examples, upon ranking the set of virtual spaces, the communication platform may further rank or otherwise prioritize the content found in such virtual spaces. The communication platform may prioritize the messages according to the techniques described above. Though it has been described that the communication platform may prioritize either the virtual spaces or the messages, in other examples, the communication platform may prioritize the virtual spaces and the messages.

In some examples, the communication platform may generate a summary based on the content. In some instances, the communication platform may generate a summary for the user based on using a machine-learning model trained to output summarized data. That is, the communication platform may input the content to machine-learning model and receive, as output, a summary of such content. A summary of the content may include one or more action items identified in the content, one or more recommendations, a summary of the content, and/or any other type of information. In some instances, the summary may be organized according to type of information and/or by virtual space. As such, the communication platform may receive, from the machine-learning model, a summary that represents the

content that is most relevant to the user across some or all virtual spaces that the user is a member of.

In some examples, the communication platform may cause the summary to be displayed via a user interface of the user. The communication platform may display or otherwise render the summary to one or more different locations in the communication platform. For instance, the communication platform may include a summary section included in a side pane and as such, the communication platform may associate the summary with the summary section. Accordingly, upon receiving a request (e.g., selection) from the user to access the summary section, the communication platform may cause the summary section virtual space to be displayed which may include the summary. In other examples, the communication platform may display the summary in a thread pane in the communication platform. In yet other examples, the communication platform may display the summary via a separate interface such as an overlay interface, popup box, and/or any other type of interface.

The following detailed description of examples references the accompanying drawings that illustrate specific examples in which the techniques can be practiced. The examples are intended to describe aspects of the systems and methods in sufficient detail to enable those skilled in the art to practice the techniques discussed herein. Other examples can be utilized and changes can be made without departing from the scope of the disclosure. The following detailed description is, therefore, not to be taken in a limiting sense. The scope of the disclosure is defined only by the appended claims, along with the full scope of equivalents to which such claims are entitled.

Group-Based Communication System

FIG. 1 illustrates an example environment **100** for performing techniques described herein. In at least one example, the example environment **100** can be associated with a communication platform that can leverage a network-based computing system to enable users of the communication platform to exchange data. In at least one example, the communication platform can be “group-based” such that the platform, and associated systems, communication channels, messages, collaborative documents, canvases, audio/video conversations, and/or other virtual spaces, have security (that can be defined by permissions) to limit access to a defined group of users. In some examples, such groups of users can be defined by group identifiers, as described above, which can be associated with common access credentials, domains, or the like. In some examples, the communication platform can be a hub, offering a secure and private virtual space to enable users to chat, meet, call, collaborate, transfer files or other data, or otherwise communicate between or among each other. As described above, each group can be associated with a workspace, enabling users associated with the group to chat, meet, call, collaborate, transfer files or other data, or otherwise communicate between or among each other in a secure and private virtual space. In some examples, members of a group, and thus workspace, can be associated with a same organization. In some examples, members of a group, and thus workspace, can be associated with different organizations (e.g., entities with different organization identifiers).

In at least one example, the example environment **100** can include one or more server computing devices (or “server(s)”) **102**. In at least one example, the server(s) **102** can include one or more servers or other types of computing devices that can be embodied in any number of ways. For

example, in the example of a server, the functional components and data can be implemented on a single server, a cluster of servers, a server farm or data center, a cloud-hosted computing service, a cloud-hosted storage service, and so forth, although other computer architectures can additionally or alternatively be used.

In at least one example, the server(s) **102** can communicate with a user computing device **104** via one or more network(s) **106**. That is, the server(s) **102** and the user computing device **104** can transmit, receive, and/or store data (e.g., content, information, or the like) using the network(s) **106**, as described herein. The user computing device **104** can be any suitable type of computing device, e.g., portable, semi-portable, semi-stationary, or stationary. Some examples of the user computing device **104** can include a tablet computing device, a smart phone, a mobile communication device, a laptop, a netbook, a desktop computing device, a terminal computing device, a wearable computing device, an augmented reality device, an Internet of Things (IOT) device, or any other computing device capable of sending communications and performing the functions according to the techniques described herein. While a single user computing device **104** is shown, in practice, the example environment **100** can include multiple (e.g., tens of, hundreds of, thousands of, millions of) user computing devices. In at least one example, user computing devices, such as the user computing device **104**, can be operable by users to, among other things, access communication services via the communication platform. A user can be an individual, a group of individuals, an employer, an enterprise, an organization, and/or the like.

The network(s) **106** can include, but are not limited to, any type of network known in the art, such as a local area network or a wide area network, the Internet, a wireless network, a cellular network, a local wireless network, Wi-Fi and/or close-range wireless communications, Bluetooth®, Bluetooth Low Energy (BLE), Near Field Communication (NFC), a wired network, or any other such network, or any combination thereof. Components used for such communications can depend at least in part upon the type of network, the environment selected, or both. Protocols for communicating over such network(s) **106** are well known and are not discussed herein in detail.

In at least one example, the server(s) **102** can include one or more processors **108**, computer-readable media **110**, one or more communication interfaces **112**, and/or input/output devices **114**.

In at least one example, each processor of the processor(s) **108** can be a single processing unit or multiple processing units, and can include single or multiple computing units or multiple processing cores. The processor(s) **108** can be implemented as one or more microprocessors, microcomputers, microcontrollers, digital signal processors, central processing units (CPUs), graphics processing units (GPUs), state machines, logic circuitries, and/or any devices that manipulate signals based on operational instructions. For example, the processor(s) **108** can be one or more hardware processors and/or logic circuits of any suitable type specifically programmed or configured to execute the algorithms and processes described herein. The processor(s) **108** can be configured to fetch and execute computer-readable instructions stored in the computer-readable media, which can program the processor(s) to perform the functions described herein.

The computer-readable media **110** can include volatile and nonvolatile memory and/or removable and non-removable media implemented in any type of technology for

storage of data, such as computer-readable instructions, data structures, program modules, or other data. Such computer-readable media **110** can include, but is not limited to, RAM, ROM, EEPROM, flash memory or other memory technology, optical storage, solid state storage, magnetic tape, magnetic disk storage, RAID storage systems, storage arrays, network attached storage, storage area networks, cloud storage, or any other medium that can be used to store the desired data and that can be accessed by a computing device. Depending on the configuration of the server(s) **102**, the computer-readable media **110** can be a type of computer-readable storage media and/or can be a tangible non-transitory media to the extent that when mentioned, non-transitory computer-readable media exclude media such as energy, carrier signals, electromagnetic waves, and signals per se.

The computer-readable media **110** can be used to store any number of functional components that are executable by the processor(s) **108**. In many implementations, these functional components comprise instructions or programs that are executable by the processor(s) **108** and that, when executed, specifically configure the processor(s) **108** to perform the actions attributed above to the server(s) **102**. Functional components stored in the computer-readable media can optionally include a messaging component **116**, an audio/video component **118**, a communication platform summarizing component **120**, an operating system **122**, and a datastore **124**.

In at least one example, the messaging component **116** can process messages between users. That is, in at least one example, the messaging component **116** can receive an outgoing message from a user computing device **104** and can send the message as an incoming message to a second user computing device **104**. The messages can include direct messages sent from an originating user to one or more specified users and/or communication channel messages sent via a communication channel from the originating user to the one or more users associated with the communication channel. Additionally, the messages can be transmitted in association with a collaborative document, canvas, or other collaborative space. In at least one example, the canvas can include a flexible canvas for curating, organizing, and sharing collections of information between users. In at least one example, the collaborative document can be associated with a document identifier (e.g., virtual space identifier, communication channel identifier, etc.) configured to enable messaging functionalities attributable to a virtual space (e.g., a communication channel) within the collaborative document. That is, the collaborative document can be treated as, and include the functionalities associated with, a virtual space, such as a communication channel. The virtual space, or communication channel, can be a data route used for exchanging data between and among systems and devices associated with the communication platform.

In at least one example, the messaging component **116** can establish a communication route between and among various user computing devices, allowing the user computing devices to communicate and share data between and among each other. In at least one example, the messaging component **116** can manage such communications and/or sharing of data. In some examples, data associated with a virtual space, such as a collaborative document, can be presented via a user interface. In addition, metadata associated with each message transmitted via the virtual space, such as a timestamp associated with the message, a sending user identifier, a recipient user identifier, a conversation identifier and/or a root object identifier (e.g., conversation associated with a

thread and/or a root object), and/or the like, can be stored in association with the virtual space.

In various examples, the messaging component **116** can receive a message transmitted in association with a virtual space (e.g., direct message instance, communication channel, canvas, collaborative document, etc.). In various examples, the messaging component **116** can identify one or more users associated with the virtual space and can cause a rendering of the message in association with instances of the virtual space on respective user computing devices **104**. In various examples, the messaging component **116** can identify the message as an update to the virtual space and, based on the identified update, can cause a notification associated with the update to be presented in association with a sidebar of user interface associated with one or more of the user(s) associated with the virtual space. For example, the messaging component **116** can receive, from a first user account, a message transmitted in association with a virtual space. In response to receiving the message (e.g., interaction data associated with an interaction of a first user with the virtual space), the messaging component **116** can identify a second user associated with the virtual space (e.g., another user that is a member of the virtual space). In some examples, the messaging component **116** can cause a notification of an update to the virtual space to be presented via a sidebar of a user interface associated with a second user account of the second user. In some examples, the messaging component **116** can cause the notification to be presented in response to a determination that the sidebar of the user interface associated with the second user account includes an affordance associated with the virtual space. In such examples, the notification can be presented in association with the affordance associated with the virtual space.

In various examples, the messaging component **116** can be configured to identify a mention or tag associated with the message transmitted in association with the virtual space. In at least one example, the mention or tag can include an @mention (or other special character) of a user identifier that is associated with the communication platform. The user identifier can include a username, real name, or other unique identifier that is associated with a particular user. In response to identifying the mention or tag of the user identifier, the messaging component **116** can cause a notification to be presented on a user interface associated with the user identifier, such as in association with an affordance associated with the virtual space in a sidebar of a user interface associated with the particular user and/or in a virtual space associated with mentions and reactions. That is, the messaging component **116** can be configured to alert a particular user that they were mentioned in a virtual space.

In at least one example, the audio/video component **118** can be configured to manage audio and/or video communications between and among users. In some examples, the audio and/or video communications can be associated with an audio and/or video conversation. In at least one example, the audio and/or video conversation can include a discrete identifier configured to uniquely identify the audio and/or video conversation. In some examples, the audio and/or video component **118** can store user identifiers associated with user accounts of members of a particular audio and/or video conversation, such as to identify user(s) with appropriate permissions to access the particular audio and/or video conversation.

In some examples, communications associated with an audio and/or video conversation ("conversation") can be synchronous and/or asynchronous. That is, the conversation can include a real-time audio and/or video conversation

between a first user and a second user during a period of time and, after the first period of time, a third user who is associated with (e.g., is a member of) the conversation can contribute to the conversation. The audio/video component **118** can be configured to store audio and/or video data associated with the conversation, such as to enable users with appropriate permissions to listen and/or view the audio and/or video data.

In some examples, the audio/video component **118** can be configured to generate a transcript of the conversation, and can store the transcript in association with the audio and/or video data. The transcript can include a textual representation of the audio and/or video data. In at least one example, the audio/video component **118** can use known speech recognition techniques to generate the transcript. In some examples, the audio/video component **118** can generate the transcript concurrently or substantially concurrently with the conversation. That is, in some examples, the audio/video component **118** can be configured to generate a textual representation of the conversation while it is being conducted. In some examples, the audio/video component **118** can generate the transcript after receiving an indication that the conversation is complete. The indication that the conversation is complete can include an indication that a host or administrator associated therewith has stopped the conversation, that a threshold number of meeting attendees have closed associated interfaces, and/or the like. That is, the audio/video component **118** can identify a completion of the conversation and, based on the completion, can generate the transcript associated therewith.

In at least one example, the audio/video component **118** can be configured to cause presentation of the transcript in association with a virtual space with which the audio and/or video conversation is associated. For example, a first user can initiate an audio and/or video conversation in association with a communication channel. The audio/video component **118** can process audio and/or video data between attendees of the audio and/or video conversation, and can generate a transcript of the audio and/or video data. In response to generating the transcript, the audio/video component **118** can cause the transcript to be published or otherwise presented via the communication channel. In at least one example, the audio/video component **118** can render one or more sections of the transcript selectable for commenting, such as to enable members of the communication channel to comment on, or further contribute to, the conversation. In some examples, the audio/video component **118** can update the transcript based on the comments.

In at least one example, the audio/video component **118** can manage one or more audio and/or video conversations in association with a virtual space associated with a group (e.g., organization, team, etc.) administrative or command center. The group administrative or command center can be referred to herein as a virtual (and/or digital) headquarters associated with the group. In at least one example, the audio/video component **118** can be configured to coordinate with the messaging component **116** and/or other components of the server(s) **102**, to transmit communications in association with other virtual spaces that are associated with the virtual headquarters. That is, the messaging component **116** can transmit data (e.g., messages, images, drawings, files, etc.) associated with one or more communication channels, direct messaging instances, collaborative documents, canvases, and/or the like, that are associated with the virtual headquarters. In some examples, the communication channel(s), direct messaging instance(s), collaborative document(s), canvas(es), and/or the like can have associated therewith one

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or more audio and/or video conversations managed by the audio/video component 118. That is, the audio and/or video conversations associated with the virtual headquarters can be further associated with, or independent of, one or more other virtual spaces of the virtual headquarters.

In at least one example, the communication platform summarizing component 120 can generate and/or display a summary of the content in multiple virtual spaces. As described herein, machine-learning models may be used to generate a summary of content recently posted across multiple virtual spaces of which a user is a member. In some examples, a communication platform summarizing component 120 may determine whether to generate and/or display a summary of the recently updated and/or posted content in a set of virtual spaces. For instance, the communication platform summarizing component 120 may generate a summary based on analyzing user activity data, message data, and/or virtual space data corresponding to a user. That is, the communication platform summarizing component 120 may determine or otherwise identify a set of virtual spaces that the user is a member of. For a pre-determined period of time, the communication platform summarizing component 120 can determine a first number of content items (e.g., messages, updates, etc.) that have been posted to the set of virtual spaces. The communication platform summarizing component 120 may also determine, over the same period of time, a second number indicating the number of the content items the user has viewed and/or accessed. Based on the first and second numbers, the communication platform summarizing component 120 may determine a third number that indicates how many of the content items the user has yet to view. Based on the third number meeting or exceeding a threshold value, the communication platform summarizing component 120 may determine that it should generate a summary for the user. In such instances, the communication platform summarizing component 120 may generate a summary of the content that has been posted to the set of virtual spaces. In such instances, the communication platform summarizing component 120 may cause the summary to be displayed via a user interface of the user. Additional details of operations that can be performed by the communication platform summarizing component 120 are described below.

In some examples, the communication platform can manage communication channels. In some examples, the communication platform can be a channel-based messaging platform, that in some examples, can be usable by group(s) of users. Users of the communication platform can communicate with other users via communication channels. A communication channel, or virtual space, can be a data route used for exchanging data between and among systems and devices associated with the communication platform. In some examples, a channel can be a virtual space where people can post messages, documents, and/or files. In some examples, access to channels can be controlled by permissions. In some examples, channels can be limited to a single organization, shared between different organizations, public, private, or special channels (e.g., hosted channels with guest accounts where guests can make posts but are prevented from performing certain actions, such as inviting other users to the channel). In some examples, some users can be invited to channels via email, channel invites, direct messages, text messages, and the like. Examples of channels and associated functionality are discussed throughout this disclosure.

In at least one example, the operating system 122 can manage the processor(s) 108, computer-readable media 110, hardware, software, etc. of the server(s) 102.

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In at least one example, the datastore 124 can be configured to store data that is accessible, manageable, and updatable. In some examples, the datastore 124 can be integrated with the server(s) 102, as shown in FIG. 1. In other examples, the datastore 124 can be located remotely from the server(s) 102 and can be accessible to the server(s) 102 and/or user device(s), such as the user device 104. The datastore 124 can comprise multiple databases, which can include user/org data 126 and/or virtual space data 128. Additional or alternative data may be stored in the data store and/or one or more other data stores.

In at least one example, the user/org data 126 can include data associated with users of the communication platform. In at least one example, the user/org data 126 can store data in user profiles (which can also be referred to as “user accounts”), which can store data associated with a user, including, but not limited to, one or more user identifiers associated with multiple, different organizations or entities with which the user is associated, one or more communication channel identifiers associated with communication channels to which the user has been granted access, one or more group identifiers for groups (or, organizations, teams, entities, or the like) with which the user is associated, an indication whether the user is an owner or manager of any communication channels, an indication whether the user has any communication channel restrictions, a plurality of messages, a plurality of emojis, a plurality of conversations, a plurality of conversation topics, an avatar, an email address, a real name (e.g., John Doe), a username (e.g., j doe), a password, a time zone, a status, a token, and the like.

In at least one example, the user/org data 126 can include permission data associated with permissions of individual users of the communication platform. In some examples, permissions can be set automatically or by an administrator of the communication platform, an employer, enterprise, organization, or other entity that utilizes the communication platform, a team leader, a group leader, or other entity that utilizes the communication platform for communicating with team members, group members, or the like, an individual user, or the like. Permissions associated with an individual user can be mapped to, or otherwise associated with, an account or profile within the user/org data 126. In some examples, permissions can indicate which users can communicate directly with other users, which channels a user is permitted to access, restrictions on individual channels, which workspaces the user is permitted to access, restrictions on individual workspaces, and the like. In at least one example, the permissions can support the communication platform by maintaining security for limiting access to a defined group of users. In some examples, such users can be defined by common access credentials, group identifiers, or the like, as described above.

In at least one example, the user/org data 126 can include data associated with one or more organizations of the communication platform. In at least one example, the user/org data 126 can store data in organization profiles, which can store data associated with an organization, including, but not limited to, one or more user identifiers associated with the organization, one or more virtual space identifiers associated with the organization (e.g., workspace identifiers, communication channel identifiers, direct message instance identifiers, collaborative document identifiers, canvas identifiers, audio/video conversation identifiers, etc.), an organization identifier associated with the organization, one or more organization identifiers associated with other organizations that are authorized for communication with the organization, and the like.

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In at least one example, the virtual space data **128** can include data associated with one or more virtual spaces associated with the communication platform. The virtual space data **128** can include textual data, audio data, video data, images, files, and/or any other type of data configured to be transmitted in association with a virtual space. Non-limiting examples of virtual spaces include workspaces, communication channels, direct messaging instances, collaborative documents, canvases, and audio and/or video conversations. In at least one example, the virtual space data can store data associated with individual virtual spaces separately, such as based on a discrete identifier associated with each virtual space. In some examples, a first virtual space can be associated with a second virtual space. In such examples, first virtual space data associated with the first virtual space can be stored in association with the second virtual space. For example, data associated with a collaborative document that is generated in association with a communication channel may be stored in association with the communication channel. For another example, data associated with an audio and/or video conversation that is conducted in association with a communication channel can be stored in association with the communication channel.

As discussed above, each virtual space of the communication platform can be assigned a discrete identifier that uniquely identifies the virtual space. In some examples, the virtual space identifier associated with the virtual space can include a physical address in the virtual space data **128** where data related to that virtual space is stored. A virtual space may be “public,” which may allow any user within an organization (e.g., associated with an organization identifier) to join and participate in the data sharing through the virtual space, or a virtual space may be “private,” which may restrict data communications in the virtual space to certain users or users having appropriate permissions to view. In some examples, a virtual space may be “shared,” which may allow users associated with different organizations (e.g., entities associated with different organization identifiers) to join and participate in the data sharing through the virtual space. Shared virtual spaces (e.g., shared channels) may be public such that they are accessible to any user of either organization, or they may be private such that they are restricted to access by certain users (e.g., users with appropriate permissions) of both organizations.

In some examples, the datastore **124** can be partitioned into discrete items of data that may be accessed and managed individually (e.g., data shards). Data shards can simplify many technical tasks, such as data retention, unfurling (e.g., detecting that message contents include a link, crawling the link’s metadata, and determining a uniform summary of the metadata), and integration settings. In some examples, data shards can be associated with organizations, groups (e.g., workspaces), communication channels, users, or the like.

In some examples, individual organizations can be associated with a database shard within the datastore **124** that stores data related to a particular organization identification. For example, a database shard may store electronic communication data associated with members of a particular organization, which enables members of that particular organization to communicate and exchange data with other members of the same organization in real time or near-real time. In this example, the organization itself can be the owner of the database shard and has control over where and how the related data is stored. In some examples, a database shard can store data related to two or more organizations (e.g., as in a shared virtual space).

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In some examples, individual groups can be associated with a database shard within the datastore **124** that stores data related to a particular group identification (e.g., workspace). For example, a database shard may store electronic communication data associated with members of a particular group, which enables members of that particular group to communicate and exchange data with other members of the same group in real time or near-real time. In this example, the group itself can be the owner of the database shard and has control over where and how the related data is stored.

In some examples, a virtual space can be associated with a database shard within the datastore **124** that stores data related to a particular virtual space identification. For example, a database shard may store electronic communication data associated with the virtual space, which enables members of that particular virtual space to communicate and exchange data with other members of the same virtual space in real time or near-real time. As discussed above, the communications via the virtual space can be synchronous and/or asynchronous. In at least one example, a group or organization can be the owner of the database shard and can control where and how the related data is stored.

In some examples, individual users can be associated with a database shard within the datastore **124** that stores data related to a particular user account. For example, a database shard may store electronic communication data associated with an individual user, which enables the user to communicate and exchange data with other users of the communication platform in real time or near-real time. In some examples, the user itself can be the owner of the database shard and has control over where and how the related data is stored.

In some examples, such as when a channel is shared between two organizations, each organization can be associated with its own encryption key. When a user associated with one organization posts a message or file to the shared channel it can be encrypted in the datastore **124** with the encryption key specific to the organization and the other organization can decrypt the message or file prior to accessing the message or file. Further, in examples where organizations are in different geographical areas, data associated with a particular organization can be stored in a location corresponding to the organization and temporarily cached at a location closer to a client (e.g., associated with the other organization) when such messages or files are to be accessed. Data can be maintained, stored, and/or deleted in the datastore **124** in accordance with a data governance policy associated with each specific organization.

The communication interface(s) **112** can include one or more interfaces and hardware components for enabling communication with various other devices (e.g., the user computing device **104**), such as over the network(s) **106** or directly. In some examples, the communication interface(s) **112** can facilitate communication via WebSockets, Application Programming Interfaces (APIs) (e.g., using API calls), Hypertext Transfer Protocols (HTTPS), etc.

The server(s) **102** can further be equipped with various input/output devices **114** (e.g., I/O devices). Such I/O devices **114** can include a display, various user interface controls (e.g., buttons, joystick, keyboard, mouse, touch screen, etc.), audio speakers, connection ports and so forth.

In at least one example, the user computing device **104** can include one or more processors **130**, computer-readable media **132**, one or more communication interfaces **134**, and input/output devices **136**.

In at least one example, each processor of the processor(s) **130** can be a single processing unit or multiple processing

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units, and can include single or multiple computing units or multiple processing cores. The processor(s) **130** can comprise any of the types of processors described above with reference to the processor(s) **108** and may be the same as or different than the processor(s) **108**.

The computer-readable media **132** can comprise any of the types of computer-readable media **132** described above with reference to the computer-readable media **110** and may be the same as or different than the computer-readable media **110**. Functional components stored in the computer-readable media can optionally include at least one application **138** and an operating system **140**.

In at least one example, the application **138** can be a mobile application, a web application, or a desktop application, which can be provided by the communication platform or which can be an otherwise dedicated application. In some examples, individual user computing devices associated with the environment **100** can have an instance or versioned instance of the application **138**, which can be downloaded from an application store, accessible via the Internet, or otherwise executable by the processor(s) **130** to perform operations as described herein. That is, the application **138** can be an access point, enabling the user computing device **104** to interact with the server(s) **102** to access and/or use communication services available via the communication platform. In at least one example, the application **138** can facilitate the exchange of data between and among various other user computing devices, for example via the server(s) **102**. In at least one example, the application **138** can present user interfaces, as described herein. In at least one example, a user can interact with the user interfaces via touch input, keyboard input, mouse input, spoken input, or any other type of input.

A non-limiting example of a user interface **142** is shown in FIG. 1. As illustrated in FIG. 1, the user interface **142** can present data associated with one or more virtual spaces, which may include one or more workspaces. That is, in some examples, the user interface **142** can integrate data from multiple workspaces into a single user interface so that the user (e.g., of the user computing device **104**) can access and/or interact with data associated with the multiple workspaces that he or she is associated with and/or otherwise communicate with other users associated with the multiple workspaces. In some examples, the user interface **142** can include a first region **144**, or pane, that includes indicator(s) (e.g., user interface element(s) or object(s)) associated with workspace(s) with which the user (e.g., account of the user) is associated. In some examples, the user interface **142** can include a second region **146**, or pane, that includes indicator(s) (e.g., user interface element(s), affordance(s), object(s), etc.) representing data associated with the workspace(s) with which the user (e.g., account of the user) is associated. In at least one example, the second region **146** can represent a sidebar of the user interface **142**.

In at least one example, the user interface **142** can include a third region **148**, or pane, that can be associated with a data feed (or, "feed") indicating messages posted to and/or actions taken with respect to one or more communication channels and/or other virtual spaces for facilitating communications (e.g., a virtual space associated with direct message communication(s), a virtual space associated with event(s) and/or action(s), etc.) as described herein. In at least one example, data associated with the third region **148** can be associated with the same or different workspaces. That is, in some examples, the third region **148** can present data associated with the same or different workspaces via an integrated feed. In some examples, the data can be organized

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and/or is sortable by workspace, time (e.g., when associated data is posted or an associated operation is otherwise performed), type of action, communication channel, user, or the like. In some examples, such data can be associated with an indication of which user (e.g., member of the communication channel) posted the message and/or performed an action. In examples where the third region **148** presents data associated with multiple workspaces, at least some data can be associated with an indication of which workspace the data is associated with. In some examples, the third region **148** may be resized or popped out as a standalone window.

In at least one example, the operating system **140** can manage the processor(s) **130**, computer-readable media **132**, hardware, software, etc. of the server(s) **102**.

The communication interface(s) **134** can include one or more interfaces and hardware components for enabling communication with various other devices (e.g., the user computing device **104**), such as over the network(s) **106** or directly. In some examples, the communication interface(s) **134** can facilitate communication via WebSockets, APIs (e.g., using API calls), HTTPs, etc.

The user computing device **104** can further be equipped with various input/output devices **136** (e.g., I/O devices). Such I/O devices **136** can include a display, various user interface controls (e.g., buttons, joystick, keyboard, mouse, touch screen, etc.), audio speakers, connection ports and so forth.

While techniques described herein are described as being performed by the messaging component **116**, the audio/video component **118**, the communication platform summarizing component **120**, and the application **138**, techniques described herein can be performed by any other component, or combination of components, which can be associated with the server(s) **102**, the user computing device **104**, or a combination thereof.

User Interface for a Group-Based Communication System

FIG. 2A illustrates a user interface **200** of a group-based communication system, which will be useful in illustrating the operation of various examples discussed herein. The group-based communication system may include communication data such as messages, queries, files, mentions, users or user profiles, interactions, tickets, channels, applications integrated into one or more channels, conversations, workspaces, or other data generated by or shared between users of the group-based communication system. In some instances, the communication data may comprise data associated with a user, such as a user identifier, channels to which the user has been granted access, groups with which the user is associated, permissions, and other user-specific information.

The user interface **200** comprises a plurality of objects such as panes, text entry fields, buttons, messages, or other user interface components that are viewable by a user of the group-based communication system. As depicted, the user interface **200** comprises a title bar **202**, a workspace pane **204**, a navigation pane **206**, channels **208**, documents **210** (e.g., collaborative documents), direct messages **212**, applications **214**, a synchronous multimedia collaboration session pane **216**, and channel pane **218**.

By way of example and without limitation, when a user opens the user interface **200** they can select a workspace via the workspace pane **204**. A particular workspace may be associated with data specific to the workspace and accessible via permissions associated with the workspace. Different

sections of the navigation pane **206** can present different data and/or options to a user. Different graphical indicators may be associated with virtual spaces (e.g., channels) to summarize an attribute of the channel (e.g., whether the channel is public, private, shared between organizations, locked, etc.). When a user selects a channel, a channel pane **218** may be presented. In some examples, the channel pane **218** can include a header, pinned items (e.g., documents or other virtual spaces), an “about” document providing an overview of the channel, and the like. In some cases, members of a channel can search within the channel, access content associated with the channel, add other members, post content, and the like. In some examples, depending on the permissions associated with a channel, users who are not members of the channel may have limited ability to interact with (or even view or otherwise access) a channel. As users navigate within a channel they can view messages **222** and may react to messages (e.g., a reaction **224**), reply in a thread, start threads, and the like. Further, a channel pane **218** can include a compose pane **228** to compose message(s) and/or other data to associate with a channel. In some examples, the user interface **200** can include a threads pane **230** that provides additional levels of detail of the messages **222**. In some examples, different panes can be resized, panes can be popped out to independent windows, and/or independent windows can be merged to multiple panes of the user interface **200**. In some examples, users may communicate with other users via a collaboration pane **216**, which may provide synchronous or asynchronous voice and/or video capabilities for communication. Of course, these are illustrative examples and additional examples of the aforementioned features are provided throughout this disclosure.

In some examples, title bar **202** comprises search bar **220**. The search bar **220** may allow users to search for content located in the current workspace of the group-based communication system, such as files, messages, channels, members, commands, functions, and the like. Users may refine their searches by attributes such as content type, content author, and by users associated with the content. Users may optionally search within specific workspaces, channels, direct message conversations, or documents. In some examples, the title bar **202** comprises navigation commands allowing a user to move backwards and forwards between different panes, as well as to view a history of accessed content. In some examples, the title bar **202** may comprise additional resources such as links to help documents and user configuration settings.

In some examples, the group-based communication system can comprise a plurality of distinct workspaces, where each workspace is associated with different groups of users and channels. Each workspace can be associated with a group identifier and one or more user identifiers can be mapped to, or otherwise associated with, the group identifier. Users corresponding to such user identifiers may be referred to as members of the group. In some examples, the user interface **200** comprises the workspace pane **204** for navigating between, adding, or deleting various workspaces in the group-based communication system. For example, a user may be a part of a workspace for Acme, where the user is an employee of or otherwise affiliated with Acme. The user may also be a member of a local volunteer organization that also uses the group-based communication system to collaborate. To navigate between the two groups, the user may use the workspace pane **204** to change from the Acme workspace to the volunteer organization workspace. A workspace may comprise one or more channels that are unique to that workspace and/or one or more channels that are shared

between one or more workspaces. For example, the Acme company may have a workspace for Acme projects, such as Project Zen, a workspace for social discussions, and an additional workspace for general company matters. In some examples, an organization, such as a particular company, may have a plurality of workspaces, and the user may be associated with one or more workspaces belonging to the organization. In yet other examples, a particular workspace can be associated with one or more organizations or other entities associated with the group-based communication system.

In some examples, the navigation pane **206** permits users to navigate between virtual spaces such as pages, channels **208**, collaborative documents **210** (such as those discussed at FIG. 2D), applications **214**, and direct messages **212** within the group-based communication system. For example, the navigation pane **206** can include indicators representing virtual spaces that can aggregate data associated with a plurality of virtual spaces of which the user is a member. In at least one example, each virtual space can be associated with an indicator in the navigation pane **206**. In some examples, an indicator can be associated with an actuation mechanism (e.g., an affordance, also referred to as a graphical element) such that when actuated, can cause the user interface **200** to present data associated with the corresponding virtual space. In at least one example, a virtual space can be associated with all unread data associated with each of the workspaces with which the user is associated. That is, in some examples, if the user requests to access the virtual space associated with “unreads,” all data that has not been read (e.g., viewed) by the user can be presented, for example in a feed. In such examples, different types of events and/or actions, which can be associated with different virtual spaces, can be presented via the same feed. In some examples, such data can be organized and/or is sortable by associated virtual space (e.g., virtual space via which the communication was transmitted), time, type of action, user, and/or the like. In some examples, such data can be associated with an indication of which user (e.g., member of the associated virtual space) posted the message and/or performed an action.

In some examples, a virtual space can be associated with the same type of event and/or action. For example, “threads” can be associated with messages, files, etc. posted in threads to messages posted in a virtual space and “mentions and reactions” can be associated with messages or threads where the user has been mentioned (e.g., via a tag) or another user has reacted (e.g., via an emoji, reaction, or the like) to a message or thread posted by the user. That is, in some examples, the same types of events and/or actions, which can be associated with different virtual spaces, can be presented via the same feed. As with the “unreads” virtual space, data associated with such virtual spaces can be organized and/or is sortable by virtual space, time, type of action, user, and/or the like.

In some examples, a virtual space can be associated with facilitating communications between a user and other users of the communication platform. For example, “connect” can be associated with enabling the user to generate invitations to communicate with one or more other users. In at least one example, responsive to receiving an indication of selection of the “connect” indicator, the communication platform can cause a connections interface to be presented.

In some examples, a virtual space can be associated with one or more boards or collaborative documents with which the user is associated. In at least one example, a document can include a collaborative document configured to be

accessed and/or edited by two or more users with appropriate permissions (e.g., viewing permissions, editing permissions, etc.). In at least one example, if the user requests to access the virtual space associated with one or more documents with which the user is associated, the one or more documents can be presented via the user interface **200**. In at least one example, the documents, as described herein, can be associated with an individual (e.g., private document for a user), a group of users (e.g., collaborative document), and/or one or more communication channels (e.g., members of the communication channel rendered access permissions to the document), such as to enable users of the communication platform to create, interact with, and/or view data associated with such documents. In some examples, the collaborative document can be a virtual space, a board, a canvas, a page, or the like for collaborative communication and/or data organization within the communication platform. In at least one example, the collaborative document can support editable text and/or objects that can be ordered, added, deleted, modified, and/or the like. In some examples, the collaborative document can be associated with permissions defining which users of a communication platform can view and/or edit the document. In some examples, a collaborative document can be associated with a communication channel, and members of the communication channel can view and/or edit the document. In some examples, a collaborative document can be sharable such that data associated with the document is accessible to and/or interactable for members of the multiple communication channels, workspaces, organizations, and/or the like.

In some examples, a virtual space can be associated with a group (e.g., organization, team, etc.) headquarters (e.g., administrative or command center). In at least one example, the group headquarters can include a virtual or digital headquarters for administrative or command functions associated with a group of users. For example, "HQ" can be associated with an interface including a list of indicators associated with virtual spaces configured to enable associated members to communicate. In at least one example, the user can associate one or more virtual spaces with the "HQ" virtual space, such as via a drag and drop operation. That is, the user can determine relevant virtual space(s) to associate with the virtual or digital headquarters, such as to associate virtual space(s) that are important to the user therewith.

In some examples, a virtual space can be associated with one or more boards or collaborative documents with which the user is associated. In at least one example, a document can include a collaborative document configured to be accessed and/or edited by two or more users with appropriate permissions (e.g., viewing permissions, editing permissions, etc.). In at least one example, if the user requests to access the virtual space associated with one or more documents with which the user is associated, the one or more documents can be presented via the user interface **200**. In at least one example, the documents, as described herein, can be associated with an individual (e.g., private document for a user), a group of users (e.g., collaborative document), and/or one or more communication channels (e.g., members of the communication channel rendered access permissions to the document), such as to enable users of the communication platform to create, interact with, and/or view data associated with such documents. In some examples, the collaborative document can be a virtual space, a board, a canvas, a page, or the like for collaborative communication and/or data organization within the communication platform. In at least one example, the collaborative document can support editable text and/or objects that can be ordered,

added, deleted, modified, and/or the like. In some examples, the collaborative document can be associated with permissions defining which users of a communication platform can view and/or edit the document. In some examples, a collaborative document can be associated with a communication channel, and members of the communication channel can view and/or edit the document. In some examples, a collaborative document can be sharable such that data associated with the document is accessible to and/or interactable for members of the multiple communication channels, workspaces, organizations, and/or the like.

Additionally or in the alternative, in some examples, a virtual space can be associated with one or more canvases with which the user is associated. In at least one example, the canvas can include a flexible canvas for curating, organizing, and sharing collections of information between users. That is, the canvas can be configured to be accessed and/or modified by two or more users with appropriate permissions. In at least one example, the canvas can be configured to enable sharing of text, images, videos, GIFs, drawings (e.g., user-generated drawing via a canvas interface), gaming content (e.g., users manipulating gaming controls synchronously or asynchronously), and/or the like. In at least one example, modifications to a canvas can include adding, deleting, and/or modifying previously shared (e.g., transmitted, presented) data. In some examples, content associated with a canvas can be shareable via another virtual space, such that data associated with the canvas is accessible to and/or rendered interactable for members of the virtual space.

The navigation pane **206** may further comprise indicators representing communication channels (e.g., the channels **208**). In some examples, the communication channels can include public channels, private channels, shared channels (e.g., between groups or organizations), single workspace channels, cross-workspace channels, combinations of the foregoing, or the like. In some examples, the communication channels represented can be associated with a single workspace. In some examples, the communication channels represented can be associated with different workspaces (e.g., cross-workspace). In at least one example, if a communication channel is cross-workspace (e.g., associated with different workspaces), the user may be associated with both workspaces, or may only be associated with one of the workspaces. In some examples, the communication channels represented can be associated with combinations of communication channels associated with a single workspace and communication channels associated with different workspaces.

In some examples, the navigation pane **206** may depict some or all of the communication channels that the user has permission to access (e.g., as determined by the permission data). In such examples, the communication channels can be arranged alphabetically, based on most recent interaction, based on frequency of interactions, based on communication channel type (e.g., public, private, shared, cross-workspace, etc.), based on workspace, in user-designated sections, or the like. In some examples, the navigation pane **206** can depict some or all of the communication channels that the user is a member of, and the user can interact with the user interface **200** to browse or view other communication channels that the user is not a member of but are not currently displayed in the navigation pane **206**. In some examples, different types of communication channels (e.g., public, private, shared, cross-workspace, etc.) can be in different sections of the navigation pane **206**, or can have their own sub-regions or sub-panes in the user interface **200**. In some examples,

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communication channels associated with different workspaces can be in different sections of the navigation pane **206**, or can have their own regions or panes in the user interface **200**.

In some examples, the indicators can be associated with graphical elements that visually differentiate types of communication channels. For example, project_zen is associated with a lock graphical element. As a non-limiting example, and for the purpose of this discussion, the lock graphical element can indicate that the associated communication channel, project_zen, is private and access thereto is limited, whereas another communication channel, general, is public and access thereto is available to any member of an organization with which the user is associated. In some examples, additional or alternative graphical elements can be used to differentiate between shared communication channels, communication channels associated with different workspaces, communication channels with which the user is or is not a current member, and/or the like.

In at least one example, the navigation pane **206** can include indicators representative of communications with individual users or multiple specified users (e.g., instead of all, or a subset of, members of an organization). Such communications can be referred to as “direct messages.” The navigation pane **206** can include indicators representative of virtual spaces that are associated with private messages between one or more users.

The direct messages **212** may be communications between a first user and a second user, or they may be multi-person direct messages between a first user and two or more second users. The navigation pane **206** may be sorted and organized into hierarchies or sections depending on the user’s preferences. In some examples, all of the channels to which a user has been granted access may appear in the navigation pane **206**. In other examples, the user may choose to hide certain channels or collapse sections containing certain channels. Items in the navigation pane **206** may indicate when a new message or update has been received or is currently unread, such as by bolding the text associated with a channel in which an unread message is located or adding an icon or badge (for example, with a count of unread messages) to the channel name. In some examples, the group-based communication system may additionally or alternatively store permissions data associated with permissions of individual users of the group-based communication system, indicating which channels a user may view or join. Permissions can indicate, for example, which users can communicate directly with other users, which channels a user is permitted to access, restrictions on individual channels, which workspaces the user is permitted to access, and restrictions on individual workspaces.

Additionally or in the alternative, the navigation pane **206** can include a sub-section that is a personalized sub-section associated with a team of which the user is a member. That is, the “team” sub-section can include affordance(s) of one or more virtual spaces that are associated with the team, such as communication channels, collaborative documents, direct messaging instances, audio or video synchronous or asynchronous meetings, and/or the like. In at least one example, the user can associate selected virtual spaces with the team sub-section, such as by dragging and dropping, pinning, or otherwise associating selected virtual spaces with the team sub-section.

Channels Within the Group-Based Communication System

In some examples, the group-based communication system is a channel-based messaging platform, as shown in

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FIG. 2A. Within the group-based communication system, communication may be organized into channels, each dedicated to a particular topic and a set of users. Channels are generally a virtual space relating to a particular topic comprising messages and files posted by members of the channel.

For purposes of this discussion, a “message” can refer to any electronically generated digital object provided by a user using the user computing device **104** and that is configured for display within a communication channel and/or other virtual space for facilitating communications (e.g., a virtual space associated with direct message communication(s), etc.) as described herein. A message may include any text, image, video, audio, or combination thereof provided by a user (using a user computing device). For instance, the user may provide a message that includes text, as well as an image and a video, within the message as message contents. In such an example, the text, image, and video would comprise the message. Each message sent or posted to a communication channel of the communication platform can include metadata comprising a sending user identifier, a message identifier, message contents, a group identifier, a communication channel identifier, or the like. In at least one example, each of the foregoing identifiers may comprise American Standard Code for Information Interchange (ASCII) text, a pointer, a memory address, or the like.

The channel discussion may persist for days, months, or years and provide a historical log of user activity. Members of a particular channel can post messages within that channel that are visible to other members of that channel together with other messages in that channel. Users may select a channel for viewing to see only those messages relevant to the topic of that channel without seeing messages posted in other channels on different topics. For example, a software development company may have different channels for each software product being developed, where developers working on each particular project can converse on a generally singular topic (e.g., project) without noise from unrelated topics. Because the channels are generally persistent and directed to a particular topic or group, users can quickly and easily refer to previous communications for reference. In some examples, the channel pane **218** may display information related to a channel that a user has selected in the navigation pane **206**. For example, a user may select the project_zen channel to discuss the ongoing software development efforts for Project Zen. In some examples, the channel pane **218** may include a header comprising information about the channel, such as the channel name, the list of users in the channel, and other channel controls. Users may be able to pin items to the header for later access and add bookmarks to the header. In some examples, links to collaborative documents may be included in the header. In further examples, each channel may have a corresponding virtual space which includes channel-related information such as a channel summary, tasks, bookmarks, pinned documents, and other channel-related links which may be editable by members of the channel.

A communication channel or other virtual space can be associated with data and/or content other than messages, or data and/or content that is associated with messages. For example, non-limiting examples of additional data that can be presented via the channel pane **218** of the user interface **200** include collaborative documents (e.g., documents that can be edited collaboratively, in real-time or near real-time, etc.), audio and/or video data associated with a conversation, members added to and/or removed from the communication

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channel, file(s) (e.g., file attachment(s)) uploaded and/or removed from the communication channel), application(s) added to and/or removed from the communication channel, post(s) (data that can be edited collaboratively, in near real-time by one or members of a communication channel) added to and/or removed from the communication channel, description added to, modified, and/or removed from the communication channel, modifications of properties of the communication channel, etc.

The channel pane **218** may include messages such as message **222**, which is content posted by a user into the channel. Users may post text, images, videos, audio, or any other file as the message **222**. In some examples, particular identifiers (in messages or otherwise) may be denoted by prefixing them with predetermined characters. For example, channels may be prefixed by the “#” character (as in #project_zen) and username may be prefixed by the “@” character (as in @J_Smith or @User_A). Messages such as the message **222** may include an indication of which user posted the message and the time at which the message was posted. In some examples, users may react to messages by selecting a reaction button **224**. The reaction button **224** allows users to select an icon (sometimes called a reacji in this context), such as a thumbs up, to be associated with the message. Users may respond to messages, such as the message **222**, of another user with a new message. In some examples, such conversations in channels may further be broken out into threads. Threads may be used to aggregate messages related to a particular conversation together to make the conversation easier to follow and reply to, without cluttering the main channel with the discussion. Under the message beginning the thread appears a thread reply preview **226**. The thread reply preview **226** may show information related to the thread, such as, for example, the number of replies and the members who have replied. Thread replies may appear in a thread pane **230** that may be separate from the channel pane **218** and may be viewed by other members of the channel by selecting the thread reply preview **226** in the channel pane **218**.

In some examples, one or both of the channel pane **218** and the thread pane **230** may include a compose pane **228**. In some examples, the compose pane **228** allows users to compose and transmit messages **222** to the members of the channel or to those members of the channel who are following the thread (when the message is sent in a thread). The compose pane **228** may have text editing functions such as bold, strikethrough, and italicize, and/or may allow users to format their messages or attach files such as collaborative documents, images, videos, or any other files to share with other members of the channel. In some examples, the compose pane **228** may enable additional formatting options such as numbered or bulleted lists via either the user interface or an API. The compose pane **228** may also function as a workflow trigger to initiate workflows related to a channel or message. In further examples, links or documents sent via the compose pane **228** may include unfurl instructions related to how the content should be displayed.

Synchronous Multimedia Collaboration Sessions

FIG. 2B illustrates a multimedia collaboration session (e.g., a synchronous multimedia collaboration session) that has been triggered from a channel, as shown in pane **216**. Synchronous multimedia collaboration sessions may provide ambient, ad hoc multimedia collaboration in the group-based communication system. Users of the group-based

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communication system can quickly and easily join and leave these synchronous multimedia collaboration sessions at any time, without disrupting the synchronous multimedia collaboration session for other users. In some examples, synchronous multimedia collaboration sessions may be based around a particular topic, a particular channel, a particular direct message or multi-person direct message, or a set of users, while in other examples, synchronous multimedia collaboration sessions may exist without being tied to any channel, topic, or set of users.

Synchronous multimedia collaboration session pane **216** may be associated with a session conducted for a plurality of users in a channel, users in a multi-person direct message conversation, or users in a direct message conversation. Thus, a synchronous multimedia collaboration session may be started for a particular channel, multi-person direct message conversation, or direct message conversation by one or more members of that channel or conversation. Users may start a synchronous multimedia collaboration session in a channel as a means of communicating with other members of that channel who are presently online. For example, a user may have an urgent decision and want immediate verbal feedback from other members of the channel. As another example, a synchronous multimedia collaboration session may be initiated with one or more other users of the group-based communication system through direct messaging. In some examples, the audience of a synchronous multimedia collaboration session may be determined based on the context in which the synchronous multimedia collaboration session was initiated. For example, starting a synchronous multimedia collaboration session in a channel may automatically invite the entire channel to attend. As another example, starting a synchronous multimedia collaboration session allows the user to start an immediate audio and/or video conversation with other members of the channel without requiring scheduling or initiating a communication session through a third-party interface. In some examples, users may be directly invited to attend a synchronous multimedia collaboration session via a message or notification.

Synchronous multimedia collaboration sessions may be short, ephemeral sessions from which no data is persisted. Alternatively, in some examples, synchronous multimedia collaboration sessions may be recorded, transcribed, and/or summarized for later review. In other examples, contents of the synchronous multimedia collaboration session may automatically be persisted in a channel associated with the synchronous multimedia collaboration session. Members of a particular synchronous multimedia collaboration session can post messages within a messaging thread associated with that synchronous multimedia collaboration session that are visible to other members of that synchronous multimedia collaboration session together with other messages in that thread.

The multimedia in a synchronous multimedia collaboration session may include collaboration tools such as any or all of audio, video, screen sharing, collaborative document editing, whiteboarding, co-programming, or any other form of media. Synchronous multimedia collaboration sessions may also permit a user to share the user's screen with other members of the synchronous multimedia collaboration session. In some examples, members of the synchronous multimedia collaboration session may mark-up, comment on, draw on, or otherwise annotate a shared screen. In further examples, such annotations may be saved and persisted after the synchronous multimedia collaboration session has

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ended. A canvas may be created directly from a synchronous multimedia collaboration session to further enhance the collaboration between users.

In some examples, a user may start a synchronous multimedia collaboration session via a toggle in synchronous multimedia collaboration session pane **216** shown in FIG. **2B**. Once a synchronous multimedia collaboration session has been started, synchronous multimedia collaboration session pane **216** may be expanded to provide information about the synchronous multimedia collaboration session such as how many members are present, which user is currently talking, which user is sharing the user's screen, and/or screen share preview **232**. In some examples, users in the synchronous multimedia collaboration session may be displayed with an icon indicating that they are participating in the synchronous multimedia collaboration session. In further examples, an expanded view of the participants may show which users are active in the synchronous multimedia collaboration session and which are not. Screen share preview **232** may depict the desktop view of a user sharing the user's screen, or a particular application or presentation. Changes to the user's screen, such as the user advancing to the next slide in a presentation, will automatically be depicted in screen share preview **232**. In some examples, the screen share preview **232** may be actuated to cause the screen share preview **232** to be enlarged such that it is displayed as its own pane within the group-based communication system. In some examples, the screen share preview **232** can be actuated to cause the screen share preview **232** to pop out into a new window or application separate and distinct from the group-based communication system. In some examples, the synchronous multimedia collaboration session pane **216** may comprise tools for the synchronous multimedia collaboration session allowing a user to mute the user's microphone or invite other users. In some examples, the synchronous multimedia collaboration session pane **216** may comprise a screen share button **234** that may permit a user to share the user's screen with other members of the synchronous multimedia collaboration session pane **216**. In some examples, the screen share button **234** may provide a user with additional controls during a screen share. For example, a user sharing the user's screen may be provided with additional screen share controls to specify which screen to share, to annotate the shared screen, or to save the shared screen.

In some cases, the synchronous multimedia collaboration session pane **216** may persist in the navigation pane **206** regardless of the state of the group-based communication system. In some examples, when no synchronous multimedia collaboration session is active and/or depending on which item is selected from the navigation pane **206**, the synchronous multimedia collaboration session pane **216** may be hidden or removed from being presented via the user interface **200**. In some instances, when the pane **216** is active, the pane **216** can be associated with a currently selected channel, direct message, or multi-person direct message such that a synchronous multimedia collaboration session may be initiated and associated with the currently selected channel, direct message, or multi-person direct message.

A list of synchronous multimedia collaboration sessions may include one or more active synchronous multimedia collaboration sessions selected for recommendation. For example, the synchronous multimedia collaboration sessions may be selected from a plurality of currently active synchronous multimedia collaboration sessions. Further, the synchronous multimedia collaboration sessions may be

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selected based in part on user interaction with the sessions or some association of the instant user with the sessions or users involved in the sessions. For example, the recommended synchronous multimedia collaboration sessions may be displayed based in part on the instant user having been invited to a respective synchronous multimedia collaboration session or having previously collaborated with the users in the recommended synchronous multimedia collaboration session. In some examples, the list of synchronous multimedia collaboration sessions further includes additional information for each respective synchronous multimedia collaboration session, such as an indication of the participating users or number of participating users, a topic for the synchronous multimedia collaboration session, and/or an indication of an associated group-based communication channel, multi-person direct message conversation, or direct message conversation.

In some examples, a list of recommended active users may include a plurality of group-based communication system users recommended based on at least one of user activity, user interaction, or other user information. For example, the list of recommended active users may be selected based on an active status of the users within the group-based communication system; historic, recent, or frequent user interaction with the instant user (such as communicating within the group-based communication channel); or similarity between the recommended users and the instant user (such as determining that a recommended user shares common membership in channels with the instant user). In some examples, machine learning techniques such as cluster analysis can be used to determine recommended users. The list of recommended active users may include status user information for each recommended user, such as whether the recommended user is active, in a meeting, idle, in a synchronous multimedia collaboration session, or offline. In some examples, the list of recommended active users further comprises a plurality of actuatable buttons corresponding to some of or all the recommended users (for example, those recommended users with a status indicating availability) that, when selected, may be configured to initiate at least one of a text-based communication session (such as a direct message conversation) or a synchronous multimedia collaboration session.

In some examples, one or more recommended asynchronous multimedia collaboration sessions or meetings can be displayed in an asynchronous meeting section. By contrast with a synchronous multimedia collaboration session (described above), an asynchronous multimedia collaboration session allows each participant to collaborate at a time convenient to them. This collaboration participation is then recorded for later consumption by other participants, who can generate additional multimedia replies. In some examples, the replies are aggregated in a multimedia thread (for example, a video thread) corresponding to the asynchronous multimedia collaboration session. For example, an asynchronous multimedia collaboration session may be used for an asynchronous meeting where a topic is posted in a message at the beginning of a meeting thread and participants of the meeting may reply by posting a message or a video response. The resulting thread then comprises any documents, video, or other files related to the asynchronous meeting. In some examples, a preview of a subset of video replies may be shown in the asynchronous collaboration session or thread. This can allow, for example, a user to jump to a relevant segment of the asynchronous multimedia collaboration session or to pick up where they left off previously.

Connecting Within the Group-Based Communication System

FIG. 2C illustrates user interface **200** displaying a connect pane **252**. The connect pane **252** may provide tools and resources for users to connect across different organizations, where each organization may have their own (normally private) instance of the group-based communication system or may not yet belong to the group-based communication system. For example, a first software company may have a joint venture with a second software company with whom they wish to collaborate on jointly developing a new software application. The connect pane **252** may enable users to determine which other users and organizations are already within the group-based communication system, and to invite those users and organizations currently outside of the group-based communication system to join.

The connect pane **252** may comprise a connect search bar **254**, recent contacts **256**, connections **258**, a create channel button **260**, and/or a start direct message button **262**. In some examples, the connect search bar **254** may permit a user to search for users within the group-based communication system. In some examples, only users from organizations that have connected with the user's organization will be shown in the search results. In other examples, users from any organization that uses the group-based communication system can be displayed. In still other examples, users from organizations that do not yet use the group-based communication can also be displayed, allowing the searching user to invite them to join the group-based communication system. In some examples, users can be searched for via their group-based communication system username or their email address. In some examples, email addresses may be suggested or autocompleted based on external sources of data such as email directories or the searching user's contact list.

In some examples, external organizations as well as individual users may be shown in response to a user search. External organizations may be matched based on an organization name or internet domain, as search results may include organizations that have not yet joined the group-based communication system (similar to searching and matching for a particular user, discussed above). External organizations may be ranked based in part on how many users from the user's organization have connected with users of the external organization. Responsive to a selection of an external organization in a search result, the searching user may be able to invite the external organization to connect via the group-based communication system.

In some examples, the recent contacts **256** may display users with whom the instant user has recently interacted. The recent contacts **256** may display the user's name, company, and/or a status indication. The recent contacts **256** may be ordered based on which contacts the instant user most frequently interacts with or based on the contacts with whom the instant user most recently interacted. In some examples each recent contact of the recent contacts **256** may be an actuable control allowing the instant user to quickly start a direct message conversation with the recent contact, invite them to a channel, or take any other appropriate user action for that recent contact.

In some examples, the connections **258** may display a list of companies (e.g., organizations) with which the user has interacted. For each company, the name of the company may be displayed along with the company's logo and an indication of how many interactions the user has had with the company, for example the number of conversations. In some examples, each connection of the connections **258** may be an

actuable control allowing the instant user to quickly invite the external organization to a shared channel, display recent connections with that external organization, or take any other appropriate organization action for that connection.

In some examples, the create channel button **260** allows a user to create a new shared channel between two different organizations. Selecting the create channel button **260** may further allow a user to name the new connect channel and enter a description for the connect channel. In some examples, the user may select one or more external organizations or one or more external users to add to the shared channel. In other examples, the user may add external organizations or external users to the shared channel after the shared channel is created. In some examples, the user may elect whether to make the connect channel private (e.g., accessible only by invitation from a current member of the private channel).

In some examples, the start direct message button **262** allows a user to quickly start a direct message (or multi-person direct message) with external users at an external organization. In some examples, the external user identifier at an external organization may be supplied by the instant user as the external user's group-based communication system username or as the external user's email address. In some examples, an analysis of the email domain of the external user's email address may affect the message between the user and the external user. For example, the external user's identifier may indicate (for example, based on an email address domain) that the user's organization and the external user's organization are already connected. In some such examples, the email address may be converted to a group-based communication system username. Alternatively, the external user's identifier may indicate that the external user's organization belongs to the group-based communication system but is not connected to the instant user's organization. In some such examples, an invitation to connect to the instant user's organization may be generated in response. As another alternative, the external user may not be a member of the group-based communication system, and an invitation to join the group-based communication system as a guest or a member may be generated in response.

Collaborative Documents

FIG. 2D illustrates user interface **200** displaying a collaboration document pane **264**. A collaborative document may be any file type, such as a PDF, video, audio, word processing document, etc., and is not limited to a word processing document or a spreadsheet. A collaborative document may be modified and edited by two or more users. A collaborative document may also be associated with different user permissions, such that based on a user's permissions for the document (or sections of the document as discussed below), the user may selectively be permitted to view, edit, or comment on the collaborative document (or sections of the collaborative document). As such, users within the set of users having access to the document may have varying permissions for viewing, editing, commenting, or otherwise interfacing with the collaborative document. In some examples, permissions can be determined and/or assigned automatically based on how document(s) are created and/or shared. In some examples, permission can be determined manually. Collaborative documents may allow users to simultaneously or asynchronously create and modify documents. Collaborative documents may integrate with the group-based communication system and can both initiate

workflows and be used to store the results of workflows, which are discussed further below with respect to FIGS. 3A and 3B.

In some examples, the user interface **200** can comprise one or more collaborative documents (or one or more links to such collaborative documents). A collaborative document (also referred to as a document or canvas) can include a flexible workspace for curating, organizing, and sharing collections of information between users. Such documents may be associated with a synchronous multimedia collaboration session, an asynchronous multimedia collaboration session, a channel, a multi-person direct message conversation, and/or a direct message conversation. Shared canvases can be configured to be accessed and/or modified by two or more users with appropriate permissions. Alternatively or in addition, a user might have one or more private documents that are not associated with any other users.

Further, such documents can be @mentioned, such that particular documents can be referred to within channels (or other virtual spaces or documents) and/or other users can be @mentioned within such a document. For example, @mentioning a user within a document can provide an indication to that user and/or can provide access to the document to the user. In some examples, tasks can be assigned to a user via an @mention and such task(s) can be populated in the pane or sidebar associated with that user.

In some examples, a channel and a collaborative document **268** can be associated such that when a comment is posted in a channel it can be populated to a document **268**, and vice versa.

In some examples, when a first user interacts with a collaborative document, the communication platform can identify a second user account associated with the collaborative document and present an affordance (e.g., a graphical element) in a sidebar (e.g., the navigation pane **206**) indicative of the interaction. Further, the second user can select the affordance and/or a notification associated with or representing the interaction to access the collaborative document, to efficiently access the document and view the update thereto.

In some examples, as one or more users interact with a collaborative document, an indication (e.g., an icon or other user interface element) can be presented via user interfaces with the collaborative document to represent such interactions. For examples, if a first instance of the document is presently open on a first user computing device of a first user, and a second instance of the document is presently open on a second user computing device of a second user, one or more presence indicators can be presented on the respective user interfaces to illustrate various interactions with the document and by which user. In some examples, a presence indicator may have attributes (e.g., appearance attributes) that indicate information about a respective user, such as, but not limited to, a permission level (e.g., edit permissions, read-only access, etc.), virtual-space membership (e.g., whether the member belongs to a virtual space associated with the document), and the manner in which the user is interacting with the document (e.g., currently editing, viewing, open but not active, etc.).

In some examples, a preview of a collaborative document can be provided. In some examples, a preview can comprise a summary of the collaborative document and/or a dynamic preview that displays a variety of content (e.g., as changing text, images, etc.) to allow a user to quickly understand the context of a document. In some examples, a preview can be based on user profile data associated with the user viewing

the preview (e.g., permissions associated with the user, content viewed, edited, created, etc. by the user), and the like.

In some examples, a collaborative document can be created independent of or in connection with a virtual space and/or a channel. A collaborative document can be posted in a channel and edited or interacted with as discussed herein, with various affordances or notifications indicating presence of users associated with documents and/or various interactions.

In some examples, a machine learning model can be used to determine a summary of contents of a channel and can create a collaborative document comprising the summary for posting in the channel. In some examples, the communication platform may identify the users within the virtual space, actions associated with the users, and other contributions to the conversation to generate the summary document. As such, the communication platform can enable users to create a document (e.g., a collaborative document) for summarizing content and events that transpired within the virtual space.

In some examples, documents can be configured to enable sharing of content including (but not limited to) text, images, videos, GIFs, drawings (e.g., user-generated drawings via a drawing interface), or gaming content. In some examples, users accessing a canvas can add new content or delete (or modify) content previously added. In some examples, appropriate permissions may be required for a user to add content or to delete or modify content added by a different user. Thus, for example, some users may only be able to access some or all of a document in view-only mode, while other users may be able to access some or all of the document in an edit mode allowing those users to add or modify its contents. In some examples, a document can be shared via a message in a channel, multi-person direct message, or direct message, such that data associated with the document is accessible to and/or rendered interactable for members of the channel or recipients of the multi-person direct message or direct message.

In some examples, the collaboration document pane **264** may comprise collaborative document toolbar **266** and collaborative document **268**. In some examples, collaborative document toolbar **266** may provide the ability to edit or format posts, as discussed herein.

In some examples, collaborative documents may comprise free-form unstructured sections and workflow-related structured sections. In some examples, unstructured sections may include areas of the document in which a user can freely modify the collaborative document without any constraints. For example, a user may be able to freely type text to explain the purpose of the document. In some examples, a user may add a workflow or a structured workflow section by typing the name of (or otherwise mentioning) the workflow. In further examples, typing the “at” sign (@), a previously selected symbol, or a predetermined special character or symbol may provide the user with a list of workflows the user can select to add to the document. For example, a user may indicate that a marketing team member needs to sign off on a proposal by typing “!Marketing Approval” to initiate a workflow that culminates in a member of the marketing team approving the proposal. Placement of an exclamation point prior to the group name of “Marketing Approval” initiates a request for a specification action, in this case routing the proposal for approval. In some examples, structured sections may include text entry, selection menus, tables, checkboxes, tasks, calendar events, or any other document section. In further examples, structured sections

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may include text entry spaces that are a part of a workflow. For example, a user may enter text into a text entry space detailing a reason for approval, and then select a submit button that will advance the workflow to the next step of the workflow. In some examples, the user may be able to add, edit, or remove structured sections of the document that make up the workflow components.

In examples, sections of the collaborative document may have individual permissions associated with them. For example, a collaborative document having sections with individual permissions may provide a first user permission to view, edit, or comment on a first section, while a second user does not have permission to view, edit, or comment on the first section. Alternatively, a first user may have permissions to view a first section of the collaborative document, while a second user has permissions to both view and edit the first section of the collaborative document. The permissions associated with a particular section of the document may be assigned by a first user via various methods, including manual selection of the particular section of the document by the first user or another user with permission to assign permissions, typing or selecting an "assignment" indicator, such as the "@" symbol, or selecting the section by a name of the section. In further examples, permissions can be assigned for a plurality of collaborative documents at a single instance via these methods. For example, a plurality of collaborative documents each has a section entitled "Group Information," where the first user with permission to assign permissions desires an entire user group to have access to the information in the "Group Information" section of the plurality of collaborative documents. In examples, the first user can select the plurality of collaborative documents and the "Group Information" section to effectuate permissions to access (or view, edit, etc.) to the entire user group the "Group Information" section of each of the plurality of collaborative documents.

Automation in the Group-Based Communication System

FIG. 3A illustrates user interface 300 for automation in the group-based communication system. Automation, also referred to as workflows, allow users to automate functionality within the group-based communication system. Workflow builder 302 is depicted which allows a user to create new workflows, modify existing workflows, and review the workflow activity. Workflow builder 302 may comprise a workflow tab 304, an activity tab 306, and/or a settings tab 308. In some examples, workflow builder may include a publish button 314 which permits a user to publish a new or modified workflow.

The workflow tab 304 may be selected to enable a user to create a new workflow or to modify an existing workflow. For example, a user may wish to create a workflow to automatically welcome new users who join a channel. A workflow may comprise workflow steps 310. Workflow steps 310 may comprise at least one trigger which initiates the workflow and at least one function which takes an action once the workflow is triggered. For example, a workflow may be triggered when a user joins a channel and a function of the workflow may be to post within the channel welcoming the new user. In some examples, workflows may be triggered from a user action, such as a user reacting to a message, joining a channel, or collaborating in a collaborative document, from a scheduled date and time, or from a web request from a third-party application or service. In further examples, workflow functionality may include send-

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ing messages or forms to users, channels, or any other virtual space, modifying collaborative documents, or interfacing with applications. Workflow functionality may include workflow variables 312. For example, a welcome message may include a user's name via a variable to allow for a customized message. Users may edit existing workflow steps or add new workflow steps depending on the desired workflow functionality. Once a workflow is complete, a user may publish the workflow using publish button 314. A published workflow will wait until it is triggered, at which point the functions will be executed.

Activity tab 306 may display information related to a workflow's activity. In some examples, the activity tab 306 may show how many times a workflow has been executed. In further examples, the activity tab 306 may include information related to each workflow execution including the status, last activity date, time of execution, user who initiated the workflow, and other relevant information. The activity tab 306 may permit a user to sort and filter the workflow activity to find useful information.

A settings tab 308 may permit a user to modify the settings of a workflow. In some examples, a user may change a title or an icon associated with the workflow. Users may also manage the collaborators associated with a workflow. For example, a user may add additional users to a workflow as collaborators such that the additional users can modify the workflow. In some examples, settings tab 308 may also permit a user to delete a workflow.

FIG. 3B depicts elements related to workflows in the group-based communication system and is referred to generally by reference numeral 316. In various examples, trigger(s) 318 can be configured to invoke execution of function(s) 336 responsive to user instructions. A trigger initiates function execution and may take the form of one or more schedule(s) 320, webhook(s) 322, shortcut(s) 324, and/or slash command(s) 326. In some examples, the schedule 320 operates like a timer so that a trigger may be scheduled to fire periodically or once at a predetermined point in the future. In some examples, an end user of an event-based application sets an arbitrary schedule for the firing of a trigger, such as once-an-hour or every day at 9:15 AM.

Additionally, triggers 318 may take the form of the webhook 322. The webhook 322 may be a software component that listens at a webhook URL and port. In some examples, a trigger fires when an appropriate HTTP request is received at the webhook URL and port. In some examples, the webhook 322 requires proper authentication such as by way of a bearer token. In other examples, triggering will be dependent on payload content.

Another source of one of the trigger(s) 318 is a shortcut in the shortcut(s) 324. In some examples, the shortcut(s) 324 may be global to a group-based communication system and are not specific to a group-based communication system channel or workspace. Global shortcuts may trigger functions that are able to execute without the context of a particular group-based communication system message or group-based communication channel. By contrast, message- or channel-based shortcuts are specific to a group-based communication system message or channel and operate in the context of the group-based communication system message or group-based communication channel.

A further source of one of triggers 318 may be provided by way of slash commands 326. In some examples, the slash command(s) 326 may serve as entry points for group-based communication system functions, integrations with external services, or group-based communication system message

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responses. In some examples, the slash commands **326** may be entered by a user of a group-based communication system to trigger execution of application functionality. Slash commands may be followed by slash-command-line parameters that may be passed along to any group-based communication system function that is invoked in connection with the triggering of a group-based communication system function such as one of functions **336**.

An additional way in which a function is invoked is when an event (such as one of events **328**) matches one or more conditions as predetermined in a subscription (such as subscription **334**). Events **328** may be subscribed to by any number of subscriptions **334**, and each subscription may specify different conditions and trigger a different function. In some examples, events are implemented as group-based communication system messages that are received in one or more group-based communication system channels. For example, all events may be posted as non-user visible messages in an associated channel, which is monitored by subscriptions **334**. App events **330** may be group-based communication system messages with associated metadata that are created by an application in a group-based communication system channel. Events **328** may also be direct messages received by one or more group-based communication system users, which may be an actual user or a technical user, such as a bot. A bot is a technical user of a group-based communication system that is used to automate tasks. A bot may be controlled programmatically to perform various functions. A bot may monitor and help process group-based communication system channel activity as well as post messages in group-based communication system channels and react to members' in-channel activity. Bots may be able to post messages and upload files as well as be invited or removed from both public and private channels in a group-based communication system.

Events **328** may also be any event associated with a group-based communication system. Such group-based communication system events **332** include events relating to the creation, modification, or deletion of a user account in a group-based communication system or events relating to messages in a group-based communication system channel, such as creating a message, editing or deleting a message, or reacting to a message. Events **328** may also relate to creation, modification, or deletion of a group-based communication system channel or the membership of a channel. Events **328** may also relate to user profile modification or group creation, member maintenance, or group deletion.

As described above, subscription **334** indicates one or more conditions that, when matched by events, trigger a function. In some examples, a set of event subscriptions is maintained in connection with a group-based communication system such that when an event occurs, information regarding the event is matched against a set of subscriptions to determine which (if any) of functions **336** should be invoked. In some examples, the events to which a particular application may subscribe are governed by an authorization framework. In some instances, the event types matched against subscriptions are governed by OAuth permission scopes that may be maintained by an administrator of a particular group-based communication system.

In some examples, functions **336** can be triggered by triggers **318** and events **328** to which the function is subscribed. Functions **336** take zero or more inputs, perform processing (potentially including accessing external resources), and return zero or more results. Functions **336** may be implemented in various forms. First, there are group-based communication system built-ins **338**, which are

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associated with the core functionality of a particular group-based communication system. Some examples include creating a group-based communication system user or channel. Second are no-code builder functions **340** that may be developed by a user of a group-based communication system user in connection with an automation user interface such as workflow builder user interface. Third, there are hosted-code functions **342** that are implemented by way of group-based communication system applications developed as software code in connection with a software development environment.

These various types of functions **336** may in turn integrate with APIs **344**. In some examples, APIs **344** are associated with third-party services that functions **336** employ to provide a custom integration between a particular third-party service and a group-based communication system. Examples of third-party service integrations include video conferencing, sales, marketing, customer service, project management, and engineering application integration. In such an example, one of the triggers **318** would be a slash command **326** that is used to trigger a hosted-code function **342**, which makes an API call to a third-party video conferencing provider by way of one of the APIs **344**. As shown in FIG. 3B, the APIs **344** may themselves also become a source of any number of triggers **318** or events **328**. Continuing the above example, successful completion of a video conference would trigger one of the functions **336** that sends a message initiating a further API call to the third-party video conference provider to download and archive a recording of the video conference and store it in a group-based communication system channel.

In addition to integrating with APIs **344**, functions **336** may persist and access data in tables **346**. In some examples, tables **346** are implemented in connection with a database environment associated with a serverless execution environment in which a particular event-based application is executing. In some instances, tables **346** may be provided in connection with a relational database environment. In other examples, tables **346** are provided in connection with a database mechanism that does not employ relational database techniques. As shown in FIG. 3B, in some examples, reading or writing certain data to one or more of tables **346**, or data in table matching predefined conditions, is itself a source of some number of triggers **318** or events **328**. For example, if tables **346** are used to maintain ticketing data in an incident-management system, then a count of open tickets exceeding a predetermined threshold may trigger a message being posted in an incident-management channel in the group-based communication system.

FIG. 4 depicts an example block diagram **400** illustrating the interactions of components of a communication platform summarizing component **406** configured to generate and/or display summarized virtual space data.

In some examples, the example block diagram **400** may be implemented with and/or in conjunction with a group-based communication system. In this example, the example block diagram **400** may include one or more client devices **402** configured to communicate with a communication platform via a communication network **404**. Additionally, the example block diagram **400** may include a summary triggering component **408** configured to determine whether to generate a summary for a user, a content identifying component **410** configured to identify content to summarize, a content summarizing component **412** configured to summarize content data, and/or a display component **414** configured to display the summarized content data to a user interface of the client device **402**.

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In some examples, the example block diagram **400** may include one or more client devices **402** configured to communicate with a communication platform. The example block diagram **400** includes a client device **402A**, a client device **402B**, and a client device **402C**. In this example, the client device **402A** may be a laptop, the client device **402B** may be a watch, and the client device **402C** may be a mobile telephone; however, in other examples the client devices **402** may be any fixed computing device, such as a personal computer or a computer workstation, or any of a variety of mobile devices, such as a portable digital assistant, mobile telephone, smartphone, laptop computer, tablet computer, wearable devices, watch, or any combination of the aforementioned devices. In this example, the client devices **402** may communicate with the message summarizing component **406** via a communication network **404**, as described in FIG. 1.

In some examples, the communicating platform summarizing component **406** may include a summary triggering component **408** configured to determine whether to generate a summary for the user. That is, upon detecting a trigger or the satisfaction of a condition, the summary triggering component **408** may generate and display a summary for the user. In some examples, the trigger or condition may be a user request and in other examples trigger may be an amount of unread relevant information meeting or exceeding a threshold amount.

In some examples, the summary triggering component **408** may determine whether a condition imposed or otherwise determined by a user has been satisfied. As described above, the communication platform may include a virtual space that enables users to modify their user profile preferences. Specifically, the user can modify and/or enable settings to instruct the communication platform to generate and display summaries according to various times and/or covering specific virtual spaces. That is, the user can instruct the communication platform to generate summaries at various frequencies such as daily, weekly, upon a number of unread posts meeting or exceeds a threshold, and/or any other type of frequency. Further, the user can instruct the communication platform which types of virtual spaces to include and/or exclude from the summary, such as channels, boards, canvases, direct messages, multiparty direct messaging instances, and/or any other type of virtual space. Additionally, the user can list one or more specific virtual spaces the user wants included or excluded from the summary. As such, the summary triggering component **408** may determine that the condition (e.g., the user requested frequency listed by the user in the summary settings) has been satisfied and as such, the summary triggering component **408** may cause the communication platform summarizing component **406** to generate and display a summary for the user.

Additionally or alternatively, the summary triggering component **408** may determine whether to generate the summary based on a user request. The communication platform may include one or more objects located at various locations within the communication platform and/or virtual space. In some examples, some objects, upon user selection (e.g., clicking the object), may be configured to cause the communication platform to generate a summary of a set of virtual spaces.

In other examples, the summary triggering component **408** may determine whether a condition set by the communication platform has been satisfied. The condition may be a threshold above which the summary triggering component **408** may cause the communication platform summarizing component **406** to generate a summary. The satisfaction of

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the condition may occur when a threshold amount of recently (and/or relevant) posted content (e.g., content items) are determined to be unread over a period of time. Alternatively or additionally, the satisfaction of the condition may occur when a total number of posted content within a virtual space and/or a set of virtual spaces meets or exceeds a threshold, a total number of unread messages within a virtual space and/or the set of virtual spaces meets or exceeds a threshold, a virtual space and/or set of virtual spaces that have not been viewed in a threshold amount of time, a message or set of messages from user(s) having certain roles, a message having a particular keyword, etc. In some instances, the summary triggering component **408** may cause the communication platform summarizing component **406** to generate and display a summary to the applicable user. In some examples, to determine whether the condition (e.g., threshold) has been satisfied, the summary triggering component **408** may analyze, for a specific user, user activity data and/or virtual space data. For example, the summary triggering component **408** may identify some or all (or a set) of virtual spaces that the user is a member of based on user profile and/or membership data. Upon identifying the set of virtual spaces, the summary triggering component **408** may determine a first number of posts (e.g., messages, content items, updates, etc.) displayed to the set of virtual spaces over a period of time. The summary triggering component **408** may then determine a second number indicating an extent of the posts the user has viewed (or read). In such instances, the summary triggering component **408** may determine a third number indicating the number of posts the user has yet to view based on comparing the first number and the second number. If the third number meets or exceeds a threshold, the summary triggering component **408** may determine that the condition is satisfied and cause the communication platform summarizing component **406** to generate and display a summary to the user. For example, the summary triggering component **408** may determine that the user is a member of three virtual spaces and the period of time is one week. In this example, the summary triggering component **408** can determine that over the one week period of time, the first virtual space displayed 15 posts, the second virtual space displayed 90 posts, and the third virtual space displayed 50 posts. As such, the first number indicating the number of posts made to the set of virtual spaces is 155. Further, the summary triggering component **408** can determine that, over the one week period of time, the user viewed 12 messages (e.g., the second number). Accordingly, the summary triggering component **408** can determine the third number by comparing 155 (e.g., the first number) and 12 (e.g., the second number) to determine that the user has yet to view 143 posts that were made across the set of virtual spaces over the last week. As such, the summary triggering component **408** can compare 143 to a threshold which may be 100. Accordingly, the summary triggering component **408** can determine that the condition (e.g., threshold) is satisfied and cause the communication platform **406** to generate and display a summary to the user.

In some examples, the communicating platform summarizing component **406** may include a content identifying component **410** configured to identify content to summarize. Since the generated summary may be unique and relevant to the particular user, the content identifying component **410** can identify content from the set of virtual spaces that is relevant and/or important for the user. The content identifying component **410** can identify relevant and/or important content by ranking or otherwise prioritizing the content. That is, the content identifying component **410** can rank the

virtual spaces with the highest ranked virtual space being the most important and/or relevant to the user and as such should be included in the summary. Conversely, the lowest ranked virtual space may be the least important and/or relevant to the user. Additionally, the content identifying component **410** can rank individual posts (e.g., messages, content items, updates, etc.) from most relevant to the particular user to least important and relevant to the user.

As noted above, the content identifying component **410** can rank some or all posts across the virtual spaces. The messages can be ranked based on a variety of factors, such as the content of the messages, a role of the user that posted the message (e.g., a message posted by an organization leader may be highly ranked, a message posted by a subordinate employee may be ranked lower than the message posted by the organization leader, etc.), a role of the user receiving the summary (e.g., certain messages can be relevant to the user based on their role), a relationship between the posting user and the summary receiving user (e.g., organization chart), interaction data between the posting user and the summary receiving user (e.g., a frequency of interactions, a recency of interactions, a purpose of the interaction, etc.), interaction data between the summary receiving user and the virtual space within which the messages are posted, and/or any other factor.

Additionally or alternatively, the content identifying component **410** can rank virtual spaces based on which virtual space (of the set of virtual spaces) is most relevant and/or important to the particular user. As such, the virtual spaces may be ranked based on a variety of factors, such as the activity level of the user in the virtual space (e.g., a higher activity level can indicate that the virtual space is of higher importance to the user), a purpose of the virtual space to the user and/or the organization the user belongs to (e.g., company-wide channel may be of high importance to the user, a meme channel may be of lesser importance to the user, etc.), a number of unread posts, content items, and/or updates in the virtual space (e.g., a higher number of unread content may be of higher importance to summarize since it may take more time for the user to “catch up”), recency data indicating a period of time since the user joined the virtual space (e.g., a virtual space the user recently joined may be highly important to summarize since there may be large amounts of content to “catch up”), and/or any other factor.

In some examples, upon ranking the set of virtual spaces, the content identifying component **410** can further rank the content (e.g., posts) found in such virtual spaces. The communication platform may rank the messages according to the techniques described above. Though it has been described that the communication platform may prioritize either the virtual spaces or the messages, in other examples, the communication platform may prioritize the virtual spaces and the content found therein. In some examples, the content identifying component **410** may send such data (e.g., content, rankings, etc.) to the content summarizing component **412**.

In some examples, the communicating platform summarizing component **406** may include a content summarizing component **412** configured to summarize content data. The content summarizing component **412** may receive the data to summarize from the content identifying component **410**. In some examples, the content summarizing component **412** may generate a summary **416** for the user based on inputting such data into a machine-learning model. The machine-learning model may be trained using previous postings and associated rankings of such postings. Thus, the machine-learning model may be trained to generate a summary **416**

that highlights and/or organizes content according to the ranking of such content. The summary **416** may include one or more action items, one or more recommendations, a summary of the content, and/or any other type of information.

In some examples, the communicating platform summarizing component **406** may include a display component **414** configured to display the summarized content data to a user interface of the client device **402**. The display component **414** can display the summary **416** at various locations within the communication platform. For instance, the communication platform may include a summary section included in a side pane and as such, the communication platform may associate the summary with the summary section. Accordingly, upon receiving a request (e.g., selection) from the user to access the summary section, the communication platform may cause the summary section virtual space to be displayed which may include the summary. In other examples, the communication platform may display the summary in a thread pane in the communication platform. In yet other examples, the communication platform may display the summary via a separate interface such as an overlay interface, popup box, and/or any other type of interface. Further, if the display component **414** determines that the user is not signed in (e.g., not active, no active session, etc.) in the communication platform, the display component **414** may display the summary **416** in various other ways, such as via email, text message, etc. In such instances, the display component **414** may display the summary **416** on the user interface of a recipient user (as identified based on the virtual space identifier, membership data, and/or recipient user identifiers, etc.), via the communication network **404**.

FIG. 5 is a pictorial flow diagram illustrating an example process **500** for determining whether to generate a summary for a user, identifying content associated with a set of virtual spaces, generating a summary representative of the content, and displaying the summary to a user interface of the user. Specifically, FIG. 5 targets ways in which to infer whether a user may benefit from a summary of their virtual spaces.

At operation **502**, the communication platform can determine that a user is a member of a set of virtual spaces. For example, box **504** illustrates membership data of a user profile corresponding to specific user in the communication platform. Further, the box **504** illustrates various virtual spaces that the user is a member of. Specifically, the box **504** includes a type of virtual space and a name of the virtual space. In this example, the box **504** indicates that the user is a member of a channel named “general”, a channel named “company_social”, a channel named “company_alert”, a direct messaging instance between the user and John, a channel named “help_tech”, and a direct messaging instance between the user and Jim. As such, the communication platform may determine that the user is a member of the virtual spaces shown in box **504**.

At operation **506**, the communication platform can determine a first amount of activity in the virtual spaces. Activity may be user posts, responses, reactions, updates, and/or any other type of activity that can occur within the virtual space. In some examples, the communication platform may count or otherwise determine a number corresponding to the activity within the set of virtual spaces as determined at operation **502**. For instance, the communication platform may determine an amount of activity that occurred in the virtual spaces shown in box **504**.

At operation **508**, the communication platform can determine a second amount of activity in the virtual spaces that the user has viewed. A user may view a message by

accessing the virtual space within which the message is posted and/or engaging with the message is some way. In some examples, the communication platform may determine the amount of content the user has viewed by analyzing user activity data. Such activity data may indicate which virtual spaces and/or messages the user has viewed.

At operation **510**, the communication platform can determine that a third amount meets or exceeds a threshold based on the first and second amounts. The communication platform can determine the third number by comparing first number determined at operation **506** and the second number determined at operation **508**. Further, the communication platform can compare the third number to a threshold to determine whether to generate a summary for the user. For example, box **512** illustrates the first, second, and third numbers, as discussed above. In this example, box **512** illustrates that the first number (e.g., the number of new messages across the set of virtual spaces) is 322, the second number (e.g., the number of messages the user has viewed) is 28, the third number (e.g., difference between the first number and the second number) is 294, and the threshold is 150. As such, based on determining that the third number (e.g., 294) meets or exceeds the threshold (e.g., 150), the communication platform may generate and display a summary to the user interface of the user.

At operation **514**, the communication platform can identify content associated with the activity in the set of virtual spaces. In some examples, the communication platform may generate a summary based on the content in the set of virtual spaces that are relevant to the user. That is, for the communication platform to summarize the content recently posted in the set of virtual spaces, the communication platform may identify data corresponding to the set of virtual spaces to summarize. In some examples, the communication platform may prioritize the content to summarize. In such instances, the communication platform may prioritize the virtual spaces or the individual messages. Additional discussion about prioritizing or ranking content may be found in FIG. 4.

At operation **516**, the communication platform can determine a summary of the content based on inputting the content to a machine-learning model. For example, box **518** illustrates a flow diagram representing the ways in which the summary is generated. The box **518** may include a message content box which may represent the content identified at operation **514**. Further, box **518** may include a machine-learning model trained to receive, as input, the message content and output the summary of such content.

At operation **520**, the communication platform can cause the summary to be displayed via a user interface to the user. For example, box **522** may illustrate a summary being displayed via a user interface. In this example, the summary may include information regarding the company social channel, the company alerts channel, and suggestions associated with the content. Of course, this is merely an example and in other examples, the summary may include more or less information about more or less virtual spaces containing different types of information. The summary can be displayed at one or more locations within the communication platform.

FIG. 6 illustrates an example user interface **600** associated with a communication platform, as described herein, for displaying a summary of recently posted data to a summary virtual space.

In some examples, the example user interface **600** may be similar or identical to the example user interfaces of FIGS. 1-3B. In some examples, the user interface **600** may include

a navigation pane **602**. As described above, the navigation pane **602** may present different data and/or options to a user. In some examples, the navigation pane **602** may include different graphical indicators which may be associated with a variety of virtual spaces, such as channels **604**, direct messaging instances **606**, threads **608**, summary **610**, etc. The user interface **600** may include any number of channels **604** which may be used to organize conversations between and amongst users according to topics. Further, the user interface **600** may include any number of direct messaging instances **606** which may be used to organize conversations between a select number of users. The user interface **600** may include a threads **608** which may be used to access responses to one or more user posts. Additionally, the user interface **600** may include a summary **610** section which may be used to organize a summary of the content recently posted to the virtual spaces that the user is a member of. In this example, when a user selects the summary **610**, a summary pane **612** may be presented. In some examples, the summary pane **612** can include access to content associated with the summary **610**. In this example, the communication platform may render the summary **610** section to the summary pane **612** of the user interface **600**.

In some examples, the user interface **600** may include a summary viewing pane **614** to view one or more summaries that have been posted by the communication platform. As shown in FIG. 6, the summary viewing pane **614** may also include a summary **616** that is organized by virtual space and/or information type. The summary **616** may include a section containing a summary of the company social virtual space that the user is a member of. Specifically, the company social section may include a summary of the discussions and/or posts received in the company social virtual space. Further, the company social summary may also include an action item for the user. In some examples, the summary **616** may also include a section containing a summary of the company alerts virtual space that the user is a member of. Specifically, the company alerts section may include a summary of the discussions and/or posts received in the company alerts virtual space. Additionally, the company alerts summary may include multiple action items or tasks for the user to complete. Lastly, the summary **616** may include a section containing suggestions or recommendations for the user. The suggestions may be identified or otherwise inferred based on inputting the content of the virtual spaces into a machine-learning model. As such, the information included in the suggestions section may be information explicitly or implicitly included in the content of the virtual spaces.

FIG. 7 illustrates an example user interface **700** associated with a communication platform, as described herein, for displaying a summary of recently posted data to a side pane.

In some examples, the example user interface **700** may be similar or identical to the example user interfaces of FIGS. 1-3B. In some examples, the user interface **700** may include a navigation pane **702**. As described above, the navigation pane **702** may present different data and/or options to a user. In some examples, the navigation pane **702** may include different graphical indicators which may be associated with a variety of virtual spaces, such as channels **704**, direct messaging instances **706**, threads **708**, etc. The user interface **700** may include any number of channels **704** which may be used to organize conversations between and amongst users according to topics. Further, the user interface **700** may include any number of direct messaging instances **706** which may be used to organize conversations between a select few users. The user interface **700** may include a threads **708**

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which may be used to access responses to one or more user posts. In this example, when a user selects the channels **704**, a channel pane **710** may be presented. In some examples, the channel pane **710** can include access to content associated with the channel. In this example, the communication platform may render the help-tech channel **712** to the channel pane **710** of the user interface **700**.

In some examples, the user interface **700** may include a message viewing pane **714** to view one or more messages that have been posted by one or more users of the help-tech channel **712**. As shown in FIG. 7, the message viewing pane **714** may include multiple messages posted by a variety of users. Additionally, the user interface **700** may include a side pane **716** that is laterally adjacent to the message viewing pane **714**. In some examples, the side pane **716** may include access to a summary generated by the communication platform specifically for the user. In some examples, the user interface **700** may include a summary viewing pane **718** to view the summary. As shown, the summary may be a summary of multiple virtual spaces of which the user (e.g., User A) is a member. In this example, the summary may include a summary regarding the company social channel and/or the company alert channel. However, this is merely an example; in other examples, the summary may include more or less summaries and/or different types of summaries from different types of virtual spaces.

FIG. 8 illustrates an example user interface **800** associated with a communication platform, as described herein, for allowing a user to determine preferences about when and/or how the communication platform can provide the user a summary of a set of virtual spaces.

In this example, the example user interface **800** can include a user preferences interface **802** (e.g., popup box, overlay box, etc.) containing options in which a user can modify the settings of their user account/profile. As shown in FIG. 8, the user preferences interface **802** may include various sections within which users can modify their account, such as notifications, sidebar, themes, messages & media, language & region, accessibility, mark as read, audio & video, connection, accounts, privacy accounts, and/or summary **804**. In this example, the summary **804** section may include options for a user to customize their user profile regarding when and how summaries are received. Specifically, the summary **804** section may include an option **806** configured to instruct the communication platform whether the user requests a summary or not. Thus, as shown, the user has indicated that they would like the communication platform to generate and display a summary to their virtual space. The summary **804** section may include an option **808** configured to instruct the communication platform the frequency at which the user is requesting the summary. As shown, the option **808** may include frequency options such as daily, weekly, based on the number of unread messages, and/or any other frequency. The summary **804** section may include an option **810** configured to instruct the communication platform which of the user's virtual spaces the user requests the platform to summarize. As shown, the user can request that the communication platform summarize channels, canvases, multi-party direct messages along with any other type of virtual space. As such, when generating the summary, the communication platform may generate the summary including the content from the virtual space types included in option **810**. The summary **804** section may include an option **812** configured to instruct the communication platform to summarize specific virtual spaces. As shown, option **812** may include a text box within which the user can input the names of one or more virtual spaces to

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include in the summary. Accordingly, when the communication platform is determining when and how to generate a summary for the user, the communication platform may generate the summary according to the requested data provided by the user in the user preferences interface **802**.

EXAMPLE CLAUSES

A system comprising: one or more processors; and one or more non-transitory computer-readable media storing computer-executable instructions that, when executed, cause the one or more processors to perform operations comprising: determining, for a user profile of a user associated with a communication platform, that the user profile is associated with a set of virtual spaces that are maintained by and that are accessible via the communication platform; determining a first number of content items associated with the set of virtual spaces; determining, based at least in part on activity data associated with the user profile, a second number of content items that the user profile has viewed within a threshold period of time; identifying, based at least in part on the first number and the second number, content associated with the content items; determining, based at least in part on a machine-learning model and the content, a summary of the content items; and causing the summary to be displayed via a user interface of the user.

B: The system of paragraph A, wherein the summary comprises at least one of: a summary of the content, a list of one or more action items associated with the content, or a recommendation for the user.

C: The system of paragraph A, wherein identifying the set of virtual spaces is based at least in part on: receiving, from the user profile, a request comprising an instruction for the communication platform to summarize a portion of the content items associated with the set of virtual spaces.

D: The system of paragraph A, wherein determining the summary of the content items is based at least in part on: determining that a portion of the content is relevant to the user, and wherein determining that the portion of the content is relevant is based at least in part on at least one of: determining a first ranking associated with the set of virtual spaces, or determining a second ranking associated with the content items.

E: The system of paragraph D, wherein determining the first ranking is based at least in part on: an activity level of the user with a virtual space of the set of virtual spaces, a first purpose of the virtual space for the user, a second purpose of the virtual space for an organization associated with the user, a fourth number of content items of the virtual space that the user has yet to view, or recency data indicating a period of time since the user became a member of the virtual space.

F: The system of paragraph D, wherein determining the second ranking is based at least in part on: a second content associated with a content item of the content items, a role of a second user that posted the content item, a relationship between the user and the second user, or interaction data between the user and the second user.

G: The system of paragraph A, wherein identifying the content is based at least in part on: determining, based at least in part on comparing the first number and the second number, that a third number meets or exceeds a threshold value, wherein the third number represents an extent of the content items the user has yet to view during the threshold period of time.

H: One or more non-transitory computer-readable media storing instructions executable by one or more processors,

wherein the instructions, when executed, cause the one or more processors to perform operations comprising: determining, for a user profile of a user associated with a communication platform, that the user profile is associated with a set of virtual spaces associated with the communication platform; determining a first number of content items associated with the set of virtual spaces; determining, based at least in part on activity data associated with the user profile, a second number of content items that the user profile has viewed within a threshold period of time; identifying, based at least in part on the first number and the second number, content associated with the content items; determining, based at least in part on the content associated with the content items, a summary of the content items; and causing the summary to be displayed via a user interface of the user.

I: The one or more non-transitory computer-readable media of paragraph H, wherein the summary comprises at least one of: a summary of the content items, a list of one or more action items associated with the content items, or a recommendation for the user.

J: The one or more non-transitory computer-readable media of paragraph H, wherein identifying the set of virtual spaces is based at least in part on: receiving, from the user profile, a request comprising an instruction for the communication platform to summarize a portion of the content items associated with the set of virtual spaces.

K: The one or more non-transitory computer-readable media of paragraph H, wherein determining the summary of the content items is based at least in part on: identifying content associated with the content items; determining that a portion of the content is relevant to the user, and wherein determining that the portion of the content is relevant is based at least in part on at least one of: determining a first ranking associated with the set of virtual spaces, or determining a second ranking associated with the content items.

L: The one or more non-transitory computer-readable media of paragraph K, wherein determining the first ranking is based at least in part on: an activity level of the user with a virtual space of the set of virtual spaces, a first purpose of the virtual space for the user, a second purpose of the virtual space for an organization associated with the user, a fourth number of content items of the virtual space that the user has yet to view, or recency data indicating a period of time since the user became a member of the virtual space.

M: The one or more non-transitory computer-readable media of paragraph K, wherein determining the second ranking is based at least in part on: a second content associated with a content item of the content items, a role of a second user that posted the content item, a relationship between the user and the second user, or interaction data between the user and the second user.

N: A method comprising: determining, for a user profile of a user associated with a communication platform, that the user profile is associated with a set of virtual spaces associated with the communication platform; determining a first number of content items associated with the set of virtual spaces; determining, based at least in part on activity data associated with the user profile, a second number of content items that the user profile has viewed within a threshold period of time; identifying, based at least in part on the first number and the second number, content associated with the content items; determining, based at least in part on the content associated with the content items, a summary of the content items; and causing the summary to be displayed via a user interface of the user.

O: The method of paragraph N, wherein the summary comprises at least one of: a summary of the content items, a list of one or more action items associated with the content items, or a recommendation for the user.

P: The method of paragraph N, wherein identifying the set of virtual spaces is based at least in part on: receiving, from the user profile, a request comprising an instruction for the communication platform to summarize a portion of the content items associated with the set of virtual spaces.

Q: The method of paragraph N, wherein determining the summary of the content items is based at least in part on: identifying content associated with the content items; determining that a portion of the content is relevant to the user, and wherein determining that the portion of the content is relevant is based at least in part on at least one of: determining a first ranking associated with the set of virtual spaces, or determining a second ranking associated with the content items.

R: The method of paragraph Q, wherein determining the first ranking is based at least in part on: an activity level of the user with a virtual space of the set of virtual spaces, a first purpose of the virtual space for the user, a second purpose of the virtual space for an organization associated with the user, a fourth number of content items of the virtual space that the user has yet to view, or recency data indicating a period of time since the user became a member of the virtual space.

S: The method of paragraph Q, wherein determining the second ranking is based at least in part on: a second content associated with a content item of the content items, a role of a second user that posted the content item, a relationship between the user and the second user, or interaction data between the user and the second user.

T: The method of paragraph N, wherein determining the summary is based at least in part on: inputting the content into a machine-learning model; and receiving, from the machine-learning model, the summary of the content items.

While the example clauses described above are described with respect to one particular implementation, it should be understood that, in the context of this document, the content of the example clauses can also be implemented via a method, device, system, a computer-readable medium, and/or another implementation. Additionally, any of examples A-T may be implemented alone or in combination with any other one or more of the examples A-T.

CONCLUSION

While one or more examples of the techniques described herein have been described, various alterations, additions, permutations and equivalents thereof are included within the scope of the techniques described herein.

In the description of examples, reference is made to the accompanying drawings that form a part hereof, which show by way of illustration specific examples of the claimed subject matter. It is to be understood that other examples can be used and that changes or alterations, such as structural changes, can be made. Such examples, changes or alterations are not necessarily departures from the scope with respect to the intended claimed subject matter. While the steps herein can be presented in a certain order, in some cases the ordering can be changed so that certain inputs are provided at different times or in a different order without changing the function of the systems and methods described. The disclosed procedures could also be executed in different orders. Additionally, various computations that are herein need not be performed in the order disclosed, and other

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examples using alternative orderings of the computations could be readily implemented. In addition to being reordered, the computations could also be decomposed into sub-computations with the same results.

What is claimed is:

1. A system comprising:
one or more processors; and
one or more non-transitory computer-readable media storing computer-executable instructions that, when executed, cause the one or more processors to perform operations comprising:
determining, for a user profile of a user associated with a communication platform, that the user profile is associated with a set of virtual spaces that are maintained by and that are accessible via the communication platform;
determining a first number of content items associated with the set of virtual spaces;
determining, based at least in part on activity data associated with the user profile, a second number of content items that the user profile has viewed within a threshold period of time;
identifying, based at least in part on the first number and the second number, content associated with the content items;
determining, based at least in part on a machine-learning model and the content, a summary of the content items; and
causing the summary to be displayed via a user interface of the user.
2. The system of claim 1, wherein the summary comprises at least one of:
a summary of the content,
a list of one or more action items associated with the content, or
a recommendation for the user.
3. The system of claim 1, wherein identifying the set of virtual spaces is based at least in part on:
receiving, from the user profile, a request comprising an instruction for the communication platform to summarize a portion of the content items associated with the set of virtual spaces.
4. The system of claim 1, wherein determining the summary of the content items is based at least in part on:
determining that a portion of the content is relevant to the user, and wherein determining that the portion of the content is relevant is based at least in part on at least one of:
determining a first ranking associated with the set of virtual spaces, or
determining a second ranking associated with the content items.
5. The system of claim 4, wherein determining the first ranking is based at least in part on:
an activity level of the user with a virtual space of the set of virtual spaces,
a first purpose of the virtual space for the user,
a second purpose of the virtual space for an organization associated with the user,
a fourth number of content items of the virtual space that the user has yet to view, or
recency data indicating a period of time since the user became a member of the virtual space.
6. The system of claim 4, wherein determining the second ranking is based at least in part on:
a second content associated with a content item of the content items,

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a role of a second user that posted the content item, a relationship between the user and the second user, or interaction data between the user and the second user.

7. The system of claim 1, wherein identifying the content is based at least in part on:
determining, based at least in part on comparing the first number and the second number, that a third number meets or exceeds a threshold value, wherein the third number represents an extent of the content items the user has yet to view during the threshold period of time.
8. One or more non-transitory computer-readable media storing instructions executable by one or more processors, wherein the instructions, when executed, cause the one or more processors to perform operations comprising:
determining, for a user profile of a user associated with a communication platform, that the user profile is associated with a set of virtual spaces associated with the communication platform;
determining a first number of content items associated with the set of virtual spaces;
determining, based at least in part on activity data associated with the user profile, a second number of content items that the user profile has viewed within a threshold period of time;
identifying, based at least in part on the first number and the second number, content associated with the content items;
determining, based at least in part on the content associated with the content items, a summary of the content items; and
causing the summary to be displayed via a user interface of the user.
9. The one or more non-transitory computer-readable media of claim 8, wherein the summary comprises at least one of:
a summary of the content items,
a list of one or more action items associated with the content items, or
a recommendation for the user.
10. The one or more non-transitory computer-readable media of claim 8, wherein identifying the set of virtual spaces is based at least in part on:
receiving, from the user profile, a request comprising an instruction for the communication platform to summarize a portion of the content items associated with the set of virtual spaces.
11. The one or more non-transitory computer-readable media of claim 8, wherein determining the summary of the content items is based at least in part on:
identifying content associated with the content items;
determining that a portion of the content is relevant to the user, and wherein determining that the portion of the content is relevant is based at least in part on at least one of:
determining a first ranking associated with the set of virtual spaces, or
determining a second ranking associated with the content items.
12. The one or more non-transitory computer-readable media of claim 11, wherein determining the first ranking is based at least in part on:
an activity level of the user with a virtual space of the set of virtual spaces,
a first purpose of the virtual space for the user,
a second purpose of the virtual space for an organization associated with the user,

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a fourth number of content items of the virtual space that the user has yet to view, or
recency data indicating a period of time since the user became a member of the virtual space.

13. The one or more non-transitory computer-readable media of claim 11, wherein determining the second ranking is based at least in part on:

a second content associated with a content item of the content items,
a role of a second user that posted the content item,
a relationship between the user and the second user, or
interaction data between the user and the second user.

14. A method comprising:

determining, for a user profile of a user associated with a communication platform, that the user profile is associated with a set of virtual spaces associated with the communication platform;

determining a first number of content items associated with the set of virtual spaces;

determining, based at least in part on activity data associated with the user profile, a second number of content items that the user profile has viewed within a threshold period of time;

identifying, based at least in part on the first number and the second number, content associated with the content items;

determining, based at least in part on the content associated with the content items, a summary of the content items; and

causing the summary to be displayed via a user interface of the user.

15. The method of claim 14, wherein the summary comprises at least one of:

a summary of the content items,

a list of one or more action items associated with the content items, or

a recommendation for the user.

16. The method of claim 14, wherein identifying the set of virtual spaces is based at least in part on:

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receiving, from the user profile, a request comprising an instruction for the communication platform to summarize a portion of the content items associated with the set of virtual spaces.

17. The method of claim 14, wherein determining the summary of the content items is based at least in part on:

identifying content associated with the content items;

determining that a portion of the content is relevant to the user, and wherein determining that the portion of the content is relevant is based at least in part on at least one of:

determining a first ranking associated with the set of virtual spaces, or

determining a second ranking associated with the content items.

18. The method of claim 17, wherein determining the first ranking is based at least in part on:

an activity level of the user with a virtual space of the set of virtual spaces,

a first purpose of the virtual space for the user,

a second purpose of the virtual space for an organization associated with the user,

a fourth number of content items of the virtual space that the user has yet to view, or

recency data indicating a period of time since the user became a member of the virtual space.

19. The method of claim 17, wherein determining the second ranking is based at least in part on:

a second content associated with a content item of the content items,

a role of a second user that posted the content item,

a relationship between the user and the second user, or

interaction data between the user and the second user.

20. The method of claim 14, wherein determining the summary is based at least in part on:

inputting the content into a machine-learning model; and
receiving, from the machine-learning model, the summary of the content items.

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